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# Dynamics-Aware Hierarchical MADRL for Aerial-Terrestrial Cooperation in Vehicular Computation Offloading

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**Abstract**—Multi-access edge computing provides an effective computation offloading paradigm for resource-constrained vehicles by deploying resources at network edges. However, high vehicular mobility and heterogeneous task demands lead to spatio-temporal load imbalances across static roadside units (RSUs). Prior works either lack responsiveness to sudden workload surges or exhibit poor robustness under sustained fluctuations. To fill this gap, we explore an aerial-terrestrial cooperative design, leveraging UAV assistance and cross-RSU task migration to enable responsive surge absorption and system-wide robustness. Nevertheless, achieving preemptive UAV position scheduling and anticipatory cross-RSU task migration remains challenging. We argue that proactive prediction of future dynamics is crucial for guiding timely and collaborative decision-making. On this basis, we propose a Dynamics-Aware Hierarchical Multi-Agent (DHMA) deep reinforcement learning approach that jointly optimizes UAV position scheduling and vehicular computation offloading. Additionally, we develop a prediction-driven observation augmentation method to enhance the perception capabilities of agents for future dynamics based on predicted workload surges and vehicular movements. Comprehensive evaluation across four simulation scenarios, together with real-world validation, demonstrates the effectiveness, robustness, and practical viability of our approach. The source codes and datasets are publicly available here.

**Index Terms**—Computation offloading, UAV position scheduling, cross-RSU task migration, hierarchical multi-agent reinforcement learning, dynamics prediction.

## 1 Introduction

With the proliferation of intelligent vehicles, various auxiliary services such as traffic navigation, autonomous driving, augmented reality, and traffic video analysis have emerged onboard, significantly enhancing transportation efficiency and user experience [1]. However, these computation-intensive services always require low completion latency to ensure quality of service, posing great challenges for task processing on resource-constrained vehicles [2]. Meanwhile, traditional cloud computing is constrained by its high transmission latency and bandwidth overhead. Therefore, multi-access edge computing (MEC) has emerged as a promising paradigm to provide computation offloading services for vehicles [3]. By deploying resources on distributed servers at network edges, such as roadside units (RSUs) in proximity to vehicles, MEC effectively alleviates onboard computational burdens, enabling faster task responses and improved service efficiency [4].

Despite considerable benefits, achieving effective computation offloading in vehicular MEC remains challenging, as high vehicular mobility and heterogeneous task demands induce pronounced spatio-temporal load imbalances across static RSUs [5]. Spatially, since RSUs are geographically anchored with fixed resources and coverage, vehicle aggregation and dispersion at specific locations can lead to local demand-capacity mismatches [6]. Some regions evolve into hot zones where the serving RSUs become saturated, while RSUs in other low-demand regions remain underutilized with spare capacity. Temporally, these imbalances vary with diurnal traffic patterns and event-driven fluctuations [7]. For instance, morning commutes shift workloads from residential areas to business centers, whereas evening return flows reverse this pattern. Both overload and underload severely compromise resource utilization and service performance.

Recent studies increasingly recognize spatio-temporal load imbalances as a central consideration in computation offloading. Various heuristic and model-based optimization methods have been employed to mitigate such imbalances through mechanisms such as vehicle-assisted multi-edge resource cooperation [8], double-queue workload scheduling [9], and matching-based inter-server offloading [10]. However, these approaches rely on carefully designed analytical models and tailored solution procedures, which may require substantial redesign or retuning to accommodate evolving conditions. By contrast, deep reinforcement learning enables agents to learn decision policies through interactions with the environment, making it suitable for adaptive decision generation under uncertainty. Representative DRL-based designs include decentralized DDPG for MEC federation resource orchestration [11], Dueling Double DQN for load-adaptive server grouping [12], and multi-agent DRL for mobility-aware task migration [7]. Nevertheless, existing approaches either lack responsiveness when confronted with workload surges triggered by unexpected traffic events, or exhibit poor robustness with limited overload resilience, resulting in queuing backlogs and inflated tail latency.

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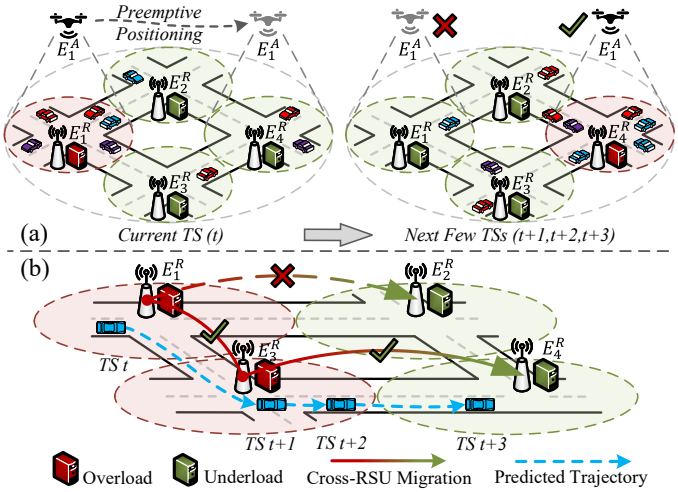



Fig. 1. (a) UAV position scheduling with workload surges. (b) Cross-RSU task migration with vehicular movements. Assuming that decisions are made every four time slots (TSs).

Consequently, designing workload-balancing mechanisms that are simultaneously responsive and robust under high vehicular dynamics remains an open and crucial problem.

Compared with RSUs, unmanned aerial vehicles (UAVs) can provide elastic resources by exploiting their high maneuverability and airborne computing capabilities [6], [13]. Deployed as aerial-level servers, UAVs can schedule their positions dynamically to deliver on-demand capacity with high LoS throughput [14], [15], thus relieving surge-induced hot zones. However, their large-scale deployment is constrained by several intrinsic limitations. Due to finite battery reserves and periodic maintenance requirements, UAVs fail to sustain long-term operations [16]. Their susceptibility to environmental variations and limited fault tolerance further compromise their stability relative to static RSUs. Accordingly, UAVs are best viewed as elastic first responders rather than persistent capacity providers. To secure system-wide robustness, lightweight and cost-effective cross-RSU task migration serves as a promising complement by redirecting excessive tasks towards underutilized RSUs. With UAVs handling surges, migration occurs between nearby RSUs with notably reduced communication costs. Building on these observations, we explore aerial-terrestrial cooperation by leveraging UAV assistance and cross-RSU task migration to ensure both responsive and robust workload balancing.

Nevertheless, achieving effective aerial-terrestrial cooperation in a UAV-assisted multi-edge collaborative computation offloading system is still nontrivial. On the one hand, dynamic workload surges exacerbate the difficulty of timely UAV positioning prior to service delivery. Hysteretic or incorrect position scheduling will leave hot zones unalleviated while aerial-level resources wasted. As shown in Figure 1(a), if RSU $E_4^R$ is expected to undergo a workload surge, the optimal positioning decision for UAV $E_1^A$ at $t$ should be $E_4^R$, despite the immediate demands on $E_1^R$. On the other hand, improper cross-RSU migration towards opposite directions of future vehicular movements can lead to increased costs [17]. Taking the vehicle in Figure 1(b) as an example, an appropriate migration decision for overloaded RSU $E_1^R$ should be $E_4^R$ rather than $E_2^R$, even though tasks initiated at $t$ need two hops to reach it. The reason is that communication costs for task transmission and result delivery continuously drop as the vehicle approaches $E_4^R$. These observations reveal two key challenges for responsive and robust workload balancing: preemptive UAV position scheduling for future workload surges and anticipatory cross-RSU task migration following future vehicular movements.

To address these challenges, we propose a **D**ynamics-Aware **H**ierarchical **M**ulti-**A**gent (DHMA) deep reinforcement learning approach. Overall, we design a two-tiered hierarchical multi-agent decision framework for effective aerial-terrestrial cooperation. At the aerial/UAV tier, UAV agents are responsible for scheduling their own positions to provide on-demand capacity in response to time-varying hot zones. To ensure preemptive position scheduling, terrestrial workload prediction is implemented to enhance the perception capabilities of UAV agents regarding sudden workload surges. At the terrestrial/RSU tier, RSU agents make computation offloading and resource management decisions for their covered vehicles, leveraging UAV-generated positioning decisions as critical inputs. To enable anticipatory direction determination of cross-RSU task migration, future vehicular movements are predicted and incorporated into the observations of RSU agents, serving as predictive guidance. In conclusion, effective aerial-terrestrial cooperation is achieved through coordinated optimization and knowledge sharing across two tiers of agents.

Our main contributions are summarized as follows:

- *UAV-Assisted Multi-Edge Collaborative Computation Offloading System.* Different from previous studies that fail to achieve responsive and robust workload balancing, we formalize computation offloading with both UAV position scheduling and cross-RSU task migration, enabling responsive surge absorption and system-wide robustness through elastic aerial assistance and terrestrial load redistribution.
- *Hierarchical Multi-Agent DRL Approach with Dynamics-Aware Observation Augmentation.* We propose a two-tiered hierarchical multi-agent decision framework together with a dynamics-aware observation augmentation method. We enhance UAV observations with future workload surges to facilitate preemptive position scheduling and incorporate future vehicular movements into RSU observations to guide anticipatory cross-RSU migration. Theoretically, we prove NP-hardness and provide convergence guarantees.
- *Performance Evaluation.* Extensive experiments are conducted on four simulation scenarios to validate the effectiveness of our approach. Moreover, comprehensive ablation studies, case studies, and real-world validation further demonstrate robustness and practical viability.

## 2 Related Work

***Computation Offloading in MEC Scenarios.*** Different computation offloading problems have been studied in existing works. Bozorgchenani et al. [18] and Wang et al. [19] focus on computation offloading in heterogeneous networks. Awada et al. [20] study resource-aware multi-task offloading and dependency-aware scheduling in an integrated edge-enabled IoV system. Jiang et al. [21] and Li et al. [22] investigate task offloading and resource allocation

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subject to energy constraints. Qiu et al. [23] optimize computation offloading and power allocation in an end-edge-cloud network. Some studies introduce UAVs into MEC systems due to their flexibility. He et al. [15] adopt one UAV as an aerial edge server to provide computing services for ground user devices. Zhou et al. [24] investigate joint service caching and task offloading in UAV-assisted wireless networks. Awada et al. [25] explore the co-scheduling of edge-equipped drones for multi-location aerial computing missions. Zhang et al. [26] optimize task offloading and UAV trajectory planning under eavesdropping threats. Tian et al. [27] tackle task offloading and UAV scheduling with explicit emphasis on user satisfaction. Guo et al. [28] address D2D association and partial offloading subject to task data dependency and UAV energy constraints. However, these studies either overlook the spatio-temporal load imbalances that commonly arise in MEC scenarios or fail to account for responsiveness and robustness simultaneously. In comparison, we propose an aerial-terrestrial cooperative design for workload balancing by coupling UAV-assisted surge absorption with cross-RSU task migration, delivering both responsiveness and system-wide robustness.

***Deep Reinforcement Learning Approaches.*** DRL-based approaches have been widely adopted to generate adaptive decisions in dynamic MEC environments, with early studies mainly employing a centralized agent. Luo et al. [29] employ DQN to choose offloading actions and flight orientations in multi-UAV cooperative target search. Zhang et al. [30] design a multipart collaborative offloading scheme based on DDPG. Wu et al. [31] integrate DQN and DDPG to optimize server selection and resource allocation in a central controller. Ren et al. [32] develop a dynamic service offloading and migration algorithm based on the asynchronous advantage actor-critic (A3C) framework. Gu et al. [33] propose a centralized SAC-based approach for partial task offloading under vehicular federated self-supervised learning. Tan et al. [34] combine Differential Evolution with Twin Delayed DDPG (TD3) to tackle uneven user distributions in multi-UAV edge computing. Li et al. [35], [36] further explore DRL for DDoS mitigation and detection in IoV security scenarios.

Recently, collaborative decision-making among a collective of agents has gained more attention. Liu et al. [37] propose a counterfactual multi-agent policy gradient (COMA)-based approach for energy-aware task migration. Zhang et al. [38] utilize COMA to optimize partial task offloading and CPU allocation. Peng et al. [39] investigate a vehicle association and resource allocation problem based on multi-agent DDPG algorithm. Zhao et al. [40] employ a multi-agent TD3 (MATD3) approach for trajectory design and task offloading of UAVs. Liu et al. [41] derive an energy-efficient offloading solution leveraging multi-agent PPO (MAPPO) algorithm. Li et al. [42] develop an adversarial MAPPO scheme for secure offloading against intelligent eavesdropping UAVs. However, existing DRL-based approaches are not directly tailored to the UAV-assisted vehicular computation offloading scenario considered in this work. Therefore, we propose a novel hierarchical multi-agent DRL approach to enable effective aerial-terrestrial cooperation through coordinated optimization and knowledge sharing across agents.

***Decision-Making with Prediction.*** Some recent studies have attempted to capture the uncertainties in MEC systems with prediction. Tang et al. [43] apply a long short-term memory (LSTM) network to model dynamic load levels at edge nodes. Guo et al. [44] employ a dual LSTM-based trajectory prediction model to direct V2V task offloading from client to server vehicles. Xu et al. [45] predict the traffic flow of different road segments with a graph weighted convolution network. Wu et al. [31] leverage the predicted coarse-grained mobility information to improve service provision. Huang et al. [11] predict the distribution of mobile devices across edge servers to facilitate the estimation of service demand. Duan et al. [12] predict the task arrival rate of all mobile devices in each MEC group based on LSTM. Chen et al. [46] build a digital twin model to predict user mobility and network resource evolution. However, prior works mainly focus on partial uncertainty prediction at coarse granularity or short horizons. To address this limitation, we design a dynamics-aware observation augmentation method that takes multi-dimensional predictions into account. We exploit the predicted workload surges and vehicular movements to guide preemptive UAV position scheduling and anticipatory cross-RSU task migration.

## 3 System Model

### 3.1 UAV-Assisted Multi-Edge Collaborative Computation Offloading System

We consider a multi-UAV multi-RSU collaborative MEC system that provides computation offloading services for intelligent vehicles with high mobility. As shown in Figure 2, there are $N_A$ UAVs $\mathcal{E}_A = \{E_1^A, E_2^A, \ldots, E_{N_A}^A\}$ and $N_R$ RSUs $\mathcal{E}_R = \{E_1^R, E_2^R, \ldots, E_{N_R}^R\}$ configured with edge servers within a multi-intersection region. Each RSU $E_i^R$ is deployed at a specific intersection and provides bandwidth $B_i^R$ for wireless communication with vehicles. Under an idealized wireless-coverage assumption, the entire road network is partitioned into non-overlapping RSU coverage regions centered at intersections. Each UAV $E_k^A$ can fly above a fixed set of intersections $\mathcal{E}_k^R$ and establish wireless connections with vehicles within one such coverage region at a time. The total available bandwidth of UAV $E_k^A$ is denoted as $B_k^A$. Note that $\mathcal{E}_R = \bigcup_{k=1}^{N_A} \mathcal{E}_k^R$, and $\mathcal{E}_k^R \bigcap \mathcal{E}_{k'}^R = \emptyset$ for $\forall k' \neq k$. Each UAV $E_k^A$ is equipped with an $n_k^A$-core CPU operating at a maximum processing frequency of $f_k^A$. Similarly, the computing power of each RSU $E_i^R$ is characterized by its $n_i^R$-core CPU with a maximum frequency of $f_i^R$. Through wireless communication, vehicles can offload their computation tasks to the RSUs or UAVs that directly cover them and get results back after execution. Moreover, data can also be transferred among RSUs through multi-hop wireless transmission, thereby enabling the migration of offloaded tasks towards other RSUs.

To simulate the continuous offloading process, we discretize time into $T$ equal-interval slots $\mathcal{T} = \{1, 2, \ldots, T\}$. At time slot $t \in \mathcal{T}$, each UAV $E_k^A$ hovers over one particular intersection in $\mathcal{E}_k^R$ and provides supplementary resource access to vehicles within its corresponding coverage region. A set of vehicles denoted as $\{v_{i,1}^t, v_{i,2}^t, \ldots, v_{i,M_i^t}^t\}$ is covered by RSU $E_i^R$, with each vehicle possessing a computing power $u_{i,j}^t$. The computation task generated on vehicle $v_{i,j}^t$ is described as $ta_{i,j}^t = \{s_{i,j}^t, c_{i,j}^t, d_{i,j}^t\}$, where $s_{i,j}^t$ indicates

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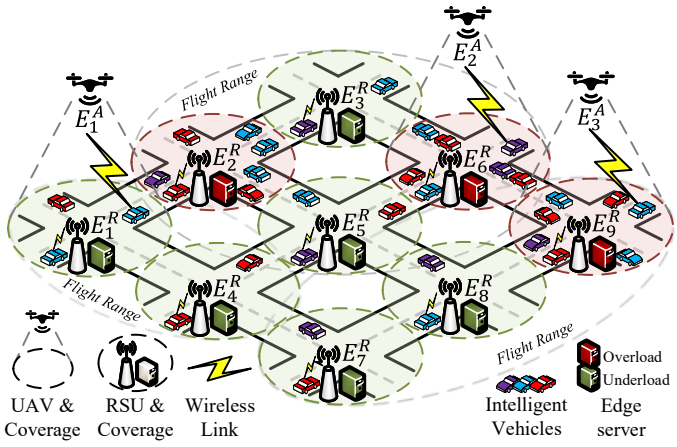



Fig. 2. Illustration of our UAV-assisted multi-edge collaborative computation offloading system within a multi-intersection region.

the data size, $c_{i,j}^t$ denotes the required CPU cycles, and $d_{i,j}^t$ specifies the maximum completion time (deadline). Each task $ta_{i,j}^t$ can be processed with onboard resources of vehicle $v_{i,j}^t$, or be offloaded to its nearest RSU $E_i^R$ for execution. Furthermore, to address the spatio-temporal load imbalances among terrestrial RSUs, the task can be strategically offloaded to an auxiliary UAV hovering directly over $E_i^R$, or migrated to a relatively idle RSU with abundant resources.

All UAVs and RSUs need to operate collaboratively to minimize the completion latency and energy consumption of tasks in the UAV-assisted multi-edge system. At time slot $t$, each UAV $E_k^A$ determines its own position by generating a one-hot decision $a_k^{s,t} = \{a_{k,x}^{s,t}\}_{x=1}^{|\mathcal{E}_k^R|}$, specifying which RSU it will fly above. If $a_{k,x}^{s,t} = 1$ and $E_i^R$ is the $x$-th RSU in $\mathcal{E}_k^R$, UAV $E_k^A$ will hover right over RSU $E_i^R$, providing aerial-level resources as supplements. Here, UAV position scheduling is modeled as slot-wise discrete hovering-point selection rather than fine-grained continuous trajectory control. After that, each RSU $E_i^R$ makes computation offloading decisions $\{a_{i,j}^{o,t}\}_{j=1}^{M_i^t}$ for its covered vehicles. We denote the $x$-th element of the one-hot decision $a_{i,j}^{o,t}$ as $a_{i,j,x}^{o,t}$, where $x \in \mathbb{N}_1^{N_R+2}$. Given $a_{i,j,x}^{o,t} = 1$, task $ta_{i,j}^t$ will be–executed locally if $x = N_R+1$; offloaded to RSU $E_i^R$ if $x = i$; offloaded to UAV $E_k^A$ if $x = N_R+2$; or migrated to an alternate RSU $E_{i'\neq i}^R$ otherwise. Considering the limited resources and heterogeneous tasks, each RSU $E_i^R$ is also responsible for performing dynamic resource management through assigning appropriate bandwidth ratios $\{a_{i,j}^{b,t}\}_{j=1}^{M_i^t}$ and processing priorities $\{a_{i,j}^{p,t}\}_{j=1}^{M_i^t}$ to its covered tasks.

### 3.2 Communication Model

To utilize the computing resources on RSUs, vehicles first need to transmit their tasks to their serving RSUs. Assuming that the wireless channel condition between vehicle $v_{i,j}^t$ and RSU $E_i^R$ remains constant, the uplink data transmission rate can be determined by the Shannon theorem:

$$r_{v\to R}^{i,j,t} = B_i^R \times a_{i,j}^{b,t} \times \log_2(1 + e_{i,j}^t \times G_i^R/(\sigma_i^R)^2), \quad (1)$$

where $e_{i,j}^t$ denotes the transmit power of vehicle $v_{i,j}^t$, $G_i^R$ is the channel gain, and $(\sigma_i^R)^2$ is the Gaussian noise power. Therefore, the vehicle-RSU transmission latency and energy consumption can be obtained by

$$l_{v\to R}^{i,j,t} = s_{i,j}^t / r_{v\to R}^{i,j,t} \text{ and } e_{v\to R}^{i,j,t} = e_{i,j}^t \times l_{v\to R}^{i,j,t}. \quad (2)$$

To responsively alleviate hot zones resulting from sudden workload surges, UAVs are flexibly deployed to deliver on-demand aerial capacity to overloaded RSUs through dynamic position scheduling. At time slot $t$, task $ta_{i,j}^t$ can be transmitted from vehicle $v_{i,j}^t$ to UAV $E_k^A$ on the condition that the UAV is hovering over RSU $E_i^R$ and has established a wireless connection with vehicle $v_{i,j}^t$. We assume that the wireless channel exhibits a similar static condition as mentioned above. Thus, task $ta_{i,j}^t$ can be offloaded to UAV $E_k^A$ with a data transmission rate of

$$r_{v\to A}^{i,j,t} = B_k^A \times a_{i,j}^{b,t} \times \log_2(1 + e_{i,j}^t \times G_k^A/(\sigma_k^A)^2), \quad (3)$$

where $G_k^A$ and $(\sigma_k^A)^2$ are the power gain and Gaussian noise power of the channel. As a result, the latency and energy consumption incurred during the vehicle-UAV transmission process can be computed as

$$l_{v\to A}^{i,j,t} = s_{i,j}^t / r_{v\to A}^{i,j,t} \text{ and } e_{v\to A}^{i,j,t} = e_{i,j}^t \times l_{v\to A}^{i,j,t}. \quad (4)$$

To further ensure system-wide robustness towards sustained fluctuations, cross-RSU task migration enables dynamic redistribution of tasks across the network, exploiting the terrestrial-level computing resources of idle RSUs. Tasks can be migrated from the current RSU $E_i^R$ to a target RSU $E_{i'\neq i}^R$ through the multi-hop wireless transmission technique. Assuming a uniform microwave transmission rate $\beta$ across all adjacent pairs of RSUs, the task migration latency and energy consumption can be calculated as

$$l_{R\to R}^{i,j,t} = s_{i,j}^t Hops(i,i')/\beta \text{ and } e_{R\to R}^{i,j,t} = e_i^R \times l_{R\to R}^{i,j,t}, \quad (5)$$

where $Hops(i,i')$ denotes the number of hops along the shortest path connecting $E_i^R$ and $E_{i'\neq i}^R$, and $e_i^R$ is the transmit power of RSU $E_i^R$.

### 3.3 Computation Model

Due to limited computing resources, arriving tasks that cannot be processed immediately will be maintained in task queues and subsequently executed following the first-in-first-out principle. The task queues on vehicles, RSUs, and UAVs are denoted as $P_{i,j}^t$, $Q_i^R$, and $Q_k^A$. Their lengths are quantified by the cumulative CPU cycles of the stored tasks.

Given the heterogeneity of vehicular tasks in both computational requirements and delay sensitivity, all tasks designated for processing on the same RSU or UAV at time slot $t$ are enqueued in strict descending order based on their assigned priority scores. More formally, for $\forall i', j', m \in \mathbb{N}_1^{N_R}$,

$$\begin{cases} ta_{i,j}^t \prec_{Q_m^R} ta_{i',j'}^t \Leftrightarrow (a_{i,j}^{p,t} > a_{i',j'}^{p,t}) \wedge (a_{i,j,m}^{o,t} \cdot a_{i',j',m}^{o,t} = 1), \\ ta_{i,j}^t \prec_{Q_k^A} ta_{i,j'}^t \Leftrightarrow (a_{i,j}^{p,t} > a_{i,j'}^{p,t}) \wedge (a_{i,j,N_R+2}^{o,t} \cdot a_{i,j',N_R+2}^{o,t} = 1), \end{cases} \quad (6)$$

where $\prec_{Q_m^R}$ and $\prec_{Q_k^A}$ denote the queue entry precedences. For each vehicle, at most one task can be assigned for local execution per time slot. Therefore, tasks are considered to enter the vehicular task queues in arrival order.

At time slot $t$, the execution of the current task $ta_{i,j}^t$ is postponed until all pending tasks in the task queue are completely cleared. This task residency process not only

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incurs waiting latency but also introduces non-negligible energy expenditure from persistent data storage. In accordance with offloading decision $a_{i,j}^{o,t}$, the waiting latency and storage-related energy consumption can be expressed as

$$\begin{cases} l_{w.v}^{i,j,t} = p_{i,j}^t / u_{i,j}^t,\ e_{s.v}^{i,j,t} = \varepsilon_{i,j}^t \times s_{i,j}^t \times l_{w.v}^{i,j,t},\ \text{if } a_{i,j,N_R+1}^{o,t} = 1, \\ l_{w.R}^{i,j,t} = q_{m,R}^{i,j,t} / f_m^R,\ e_{s.R}^{i,j,t} = \varepsilon_m^R \times s_{i,j}^t \times l_{w.R}^{i,j,t},\ \text{if } a_{i,j,m}^{o,t} = 1, \\ l_{w.A}^{i,j,t} = q_{k,A}^{i,j,t} / f_k^A,\ e_{s.A}^{i,j,t} = \varepsilon_k^A \times s_{i,j}^t \times l_{w.A}^{i,j,t},\ \text{if } a_{i,j,N_R+2}^{o,t} = 1, \end{cases} \tag{7}$$

where $p_{i,j}^t$, $q_{m,R}^{i,j,t}$, and $q_{k,A}^{i,j,t}$ denote the lengths of unprocessed tasks before task $ta_{i,j}^t$ in queues $P_{i,j}^t$, $Q_m^R$, and $Q_k^A$. While $\varepsilon_{i,j}^t$, $\varepsilon_m^R$, and $\varepsilon_k^A$ denote the energy consumption coefficients for data storage on different devices.

Task $ta_{i,j}^t$ can finally be executed after the completion of all previously enqueued tasks. In the case of local execution, the associated latency and energy consumption are given by

$$l_{e.v}^{i,j,t} = c_{i,j}^t / u_{i,j}^t \text{ and } e_{e.v}^{i,j,t} = \kappa_{i,j}^t \times (u_{i,j}^t)^2 \times c_{i,j}^t, \tag{8}$$

where $\kappa$ is the effective switched capacitance coefficient with respect to the chip architecture. Under the offloading and migration decisions towards RSU $E_m^R$, task $ta_{i,j}^t$ is executed with the latency and energy of

$$l_{e.R}^{i,j,t} = c_{i,j}^t / (f_m^R \times n_m^R) \text{ and } e_{e.R}^{i,j,t} = \kappa_m^R \times (f_m^R)^2 \times c_{i,j}^t. \tag{9}$$

Here, we assume that a task can be fully parallelized across multiple cores and the overhead of multi-core task scheduling is negligible. Thus, the all-core sustained frequency of each core under full load equals its single-core maximum frequency. Similarly, the execution latency and energy consumption under the offloading decision towards UAV $E_k^A$ can be computed as

$$l_{e.A}^{i,j,t} = c_{i,j}^t / (f_k^A \times n_k^A) \text{ and } e_{e.A}^{i,j,t} = \kappa_k^A \times (f_k^A)^2 \times c_{i,j}^t. \tag{10}$$

### 3.4 Problem Formulation

In this work, we propose a multi-objective joint optimization problem, which simultaneously considers UAV position scheduling, computation offloading, and resource management. To improve the quality of service, we focus on minimizing both the completion latency and energy consumption of all vehicular tasks in the system. Note that UAV flight time and propulsion energy are not explicitly included in the current objective, since UAV mobility is abstracted as slot-wise selection among predefined hovering points rather than continuous trajectory optimization.

If task $ta_{i,j}^t$ is executed by vehicle itself, its completion latency and energy consumption can be expressed as

$$l_{loc}^{i,j,t} = l_{w.v}^{i,j,t} + l_{e.v}^{i,j,t} \text{ and } e_{loc}^{i,j,t} = e_{s.v}^{i,j,t} + e_{e.v}^{i,j,t}. \tag{11}$$

When the task is offloaded to its nearest RSU, the transmission process must be taken into account. Thus, we get

$$l_{off.R}^{i,j,t} = l_{v\to R}^{i,j,t} + l_{w.R}^{i,j,t} + l_{e.R}^{i,j,t}, \tag{12}$$

$$e_{off.R}^{i,j,t} = e_{v\to R}^{i,j,t} + e_{s.R}^{i,j,t} + e_{e.R}^{i,j,t}. \tag{13}$$

Alternatively, to mitigate the workload surge of RSU $E_i^R$, the task is offloaded to UAV $E_k^A$ provided that the UAV is strategically deployed above RSU $E_i^R$ at time slot $t$:

$$l_{off.A}^{i,j,t} = l_{v\to A}^{i,j,t} + l_{w.A}^{i,j,t} + l_{e.A}^{i,j,t}, \tag{14}$$

$$e_{off.A}^{i,j,t} = e_{v\to A}^{i,j,t} + e_{s.A}^{i,j,t} + e_{e.A}^{i,j,t}. \tag{15}$$

TABLE 1
Notations and Descriptions.

| Notations | Descriptions |
|---|---|
| $\mathcal{E}_A/\mathcal{E}_R$ | Set of $N_A$ UAVs/$N_R$ RSUs equipped with edge servers. |
| $\mathcal{T}$ | Set of $T$ consecutive time slots. |
| $E_k^A/E_i^R$ | The $k$-th UAV/$i$-th RSU. |
| $B_k^A/B_i^R$ | Channel bandwidth of UAV $E_k^A$/RSU $E_i^R$. |
| $G_k^A/G_i^R$ | Channel power gain of UAV $E_k^A$/RSU $E_i^R$. |
| $(\sigma_k^A)^2/(\sigma_i^R)^2$ | Channel noise power of UAV $E_k^A$/RSU $E_i^R$. |
| $\beta$ | Data transmission rate between adjacent RSUs. |
| $e_i^R$ | Transmit power of RSU $E_i^R$. |
| $f_k^A/f_i^R$ | Max. CPU processing frequency of UAV $E_k^A$/RSU $E_i^R$. |
| $n_k^A/n_i^R$ | Number of CPU cores on UAV $E_k^A$/RSU $E_i^R$. |
| $\mathcal{E}_k^R$ | Set of intersections UAV $E_k^A$ can fly above. |
| $M_i^t$ | Number of vehicles covered by RSU $E_i^R$ at time slot $t$. |
| $Q_k^A/Q_i^R$ | Task queue of UAV $E_k^A$/RSU $E_i^R$. |
| $q_{k,t}^A/q_{i,t}^R$ | Length of task queue $Q_k^A/Q_i^R$ at time slot $t$. |
| $v_{i,j}^t$ | The $j$-th vehicle covered by RSU $E_i^R$ at time slot $t$. |
| $ta_{i,j}^t$ | Task of vehicle $v_{i,j}^t$. |
| $s_{i,j}^t, c_{i,j}^t, d_{i,j}^t$ | Data size, required CPU cycles, deadline of task $ta_{i,j}^t$. |
| $u_{i,j}^t$ | Max. CPU processing frequency of vehicle $v_{i,j}^t$. |
| $e_{i,j}^t$ | Transmit power of vehicle $v_{i,j}^t$. |
| $P_{i,j}^t$ | Task queue of vehicle $v_{i,j}^t$. |
| $p_{i,j}^t$ | Length of task queue $P_{i,j}^t$. |
| $a_k^{s,t}$ | Position scheduling decision for UAV $E_k^A$. |
| $a_{i,j}^{o,t}, a_{i,j}^{b,t}, a_{i,j}^{p,t}$ | Offloading, allocation, priority decisions for task $ta_{i,j}^t$. |
| $l_{loc}^{i,j,t}/e_{loc}^{i,j,t}$ | Costs of task $ta_{i,j}^t$ during local execution. |
| $l_{off.R}^{i,j,t}/e_{off.R}^{i,j,t}$ | Costs of task $ta_{i,j}^t$ if offloaded to RSU $E_i^R$. |
| $l_{off.A}^{i,j,t}/e_{off.A}^{i,j,t}$ | Costs of task $ta_{i,j}^t$ if offloaded to UAV $E_k^A$. |
| $l_{mig}^{i,j,t}/e_{mig}^{i,j,t}$ | Costs of task $ta_{i,j}^t$ if migrated to RSU $E_{i'\neq i}^R$. |
| $L^t/E^t$ | Total task completion latency/energy consumption at $t$. |

Otherwise, the task will be migrated to an idle RSU to further achieve system-wide workload balancing. The costs of the cross-RSU migration process should be considered:

$$l_{mig}^{i,j,t} = l_{v\to R}^{i,j,t} + l_{R\to R}^{i,j,t} + l_{w.R}^{i,j,t} + l_{e.R}^{i,j,t}, \tag{16}$$

$$e_{mig}^{i,j,t} = e_{v\to R}^{i,j,t} + e_{R\to R}^{i,j,t} + e_{s.R}^{i,j,t} + e_{e.R}^{i,j,t}. \tag{17}$$

We use $L^t$ and $E^t$ to denote the total latency and energy consumption for completing all tasks generated on vehicles within the UAV-assisted multi-edge collaborative computation offloading system at slot $t$. Thus, $L^t$ is represented as

$$\begin{aligned} L^t = \sum_{i=1}^{N_R} \sum_{j=1}^{M_i^t} \{ \sum_{x=1}^{N_R} \{ a_{i,j,x=i}^{o,t} l_{off.R}^{i,j,t} + a_{i,j,x\neq i}^{o,t} l_{mig}^{i,j,t} \} + \\ a_{i,j,N_R+1}^{o,t} l_{loc}^{i,j,t} + a_{i,j,N_R+2}^{o,t} l_{off.A}^{i,j,t} \}. \end{aligned} \tag{18}$$

Similarly, $E^t$ can be expressed by substituting the latency term for each computation offloading decision with its corresponding task-serving energy consumption. Consequently, the UAV-assisted multi-edge collaborative computation offloading problem throughout $T$ successive time slots can be formulated as

$$\mathcal{P}: \min \sum_{t=1}^{T} w_l L^t + w_e E^t, \tag{19}$$

$$s.t. \quad t \in \mathcal{T} = \{1, \ldots, T\}, \tag{20}$$

$$w_l, w_e \in (0,1),\ w_l + w_e = 1, \tag{21}$$

$$a_{k,x}^{s,t} \in \{0,1\},\ \sum\nolimits_{x=1}^{|\mathcal{E}_k^R|} a_{k,x}^{s,t} = 1,\ \forall k, \forall x, \tag{22}$$

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$$a_{i,j,x}^{o,t} \in \{0,1\},\ \sum_{x=1}^{N_R+2} a_{i,j,x}^{o,t} = 1,\ \forall i, \forall j, \forall x, \tag{23}$$

$$a_{i,j}^{b,t} \in (0,1],\ a_{i,j}^{p,t} \in \mathbb{R},\ \forall i, \forall j, \tag{24}$$

$$\sum_{j=1}^{M_i^t} \sum_{x=1}^{N_R} a_{i,j,x}^{o,t} \cdot a_{i,j}^{b,t} \leq 1,\ \forall i, \tag{25}$$

$$\sum_{E_i^R \in \mathcal{E}_k^R} \sum_{j=1}^{M_i^t} a_{i,j,N_R+2}^{o,t} \cdot a_{i,j}^{b,t} \leq 1,\ \forall k. \tag{26}$$

The constraints of $\mathcal{P}$ are elucidated as follows. Constraint (20) shows that the continuous offloading process $\mathcal{T}$ consists of $T$ time slots. Constraint (21) specifies the balance coefficients $w_l$ and $w_e$, which quantify the relative importance of latency and energy in the optimization objective. Constraint (22) restricts that each UAV $E_k^A$ can hover over only one RSU in $\mathcal{E}_k^R$ and communicate with vehicles within a single coverage region at a time. Constraint (23) indicates that each task is indivisible and can only be processed by one device. Constraint (24) specifies that the bandwidth ratio allocated to each task must not exceed one, and the assigned priority score takes real values. Constraints (25-26) limit the total bandwidth that can be allocated to the tasks transmitted to each RSU $E_i^R$ and UAV $E_k^A$, respectively.

***Theorem 1 (NP-Hardness).*** *The decision version $\mathcal{P}_d$ of the formulated optimization problem $\mathcal{P}$ is NP-hard. Since $\mathcal{P}_d \in$ NP, it follows that $\mathcal{P}_d$ is NP-complete. Consequently, the original optimization problem $\mathcal{P}$ can be proved to be NP-hard.*

*Proof.* The NP-hardness of $\mathcal{P}_d$ can be established by a polynomial-time reduction from the 0-1 Knapsack problem to a restricted instance of $\mathcal{P}_d$. Given any candidate solution, the feasibility of all constraints and the objective threshold can be verified in polynomial time, which implies that $\mathcal{P}_d \in$ NP. Therefore, $\mathcal{P}_d$ is NP-complete. The detailed construction and complete proof are provided in Appendix A.

***Modeling Assumptions and Applicability.*** The formulated problem assumes idealized coverage regions, simplified wireless channels, predefined UAV hovering points, fully parallelizable tasks, and non-dominant UAV mobility costs. Although these assumptions may lead to optimistic latency and energy estimates, the time-slotted formulation remains suitable for investigating the joint optimization of UAV position scheduling and vehicular computation offloading to mitigate spatio-temporal load imbalances arising in highly dynamic vehicular MEC systems.

## 4 Dynamics-Aware Hierarchical Multi-Agent Deep Reinforcement Learning

Previous analysis reveals that the multi-objective joint optimization problem $\mathcal{P}$ constitutes an NP-hard sequential decision process in high-dimensional dynamic environments, which fundamentally requires an adaptive, online learning-based resolution paradigm. To this end, we propose a Dynamics-Aware Hierarchical Multi-Agent (DHMA) deep reinforcement learning approach, as shown in Figure 4.

### 4.1 Overview

Given that computation offloading and resource management are contingent upon UAV position scheduling, directly generating all decisions with a monolithic policy would require optimizing over a large, heterogeneous joint action space containing conditionally feasible or potentially incompatible action combinations. Moreover, the two decision categories operate at different scopes and granularities, as UAV deployment determines regional aerial-resource provisioning, whereas computation offloading and resource management govern task-level services for individual vehicles. Motivated by these considerations, we decompose the formulated problem $\mathcal{P}$ into two coupled hierarchical subproblems and strategically employ two tiers of agents to solve them in a coordinated manner. As shown in Figure 3, the designed Dynamics-Aware Hierarchical Multi-Agent (DHMA) deep reinforcement learning approach consists of a UAV/aerial tier and an RSU/terrestrial tier, where UAVs and RSUs are regarded as individual agents with limited information perception scopes at their corresponding tiers. Considering the diverse traffic patterns observed throughout various intersections and regions, a centralized training and decentralized execution paradigm is adopted, enabling cooperative yet localized decision-making, in contrast to fully centralized or decentralized approaches. Adhering to this paradigm, each tier of agents leverages specific global features for cooperative training while making independent decisions with their partial observations and local policies.

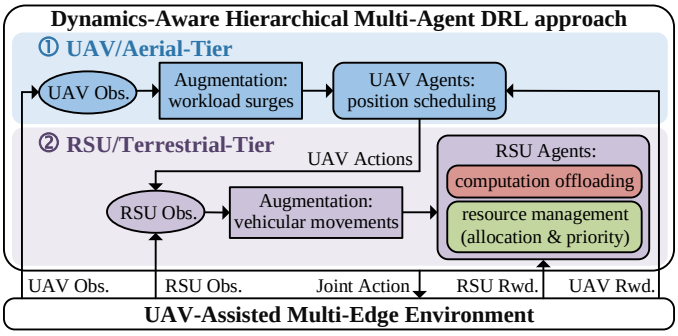



Fig. 3. Overview of our Dynamics-Aware Hierarchical Multi-Agent (DHMA) deep reinforcement learning approach.

At the UAV tier, UAV agents make position scheduling decisions according to their observations, dynamically navigating themselves towards congested intersections and delivering supplementary resources to overloaded RSUs within their flight ranges. With shared observations, multiple UAV agents can collaborate to perceive and manage large-scale load aggregations, such as traffic congestion on arterial roads that traverse the network. Furthermore, to ensure the timely delivery of aerial-level computing services, we augment the observations of UAV agents by anticipating sudden workload surges, enabling preemptive UAV positioning towards potential hot zones. Based on the determined UAV positions, RSU agents generate computation offloading and resource management decisions for their covered vehicles at the RSU tier, directing tasks to suitable devices and assigning appropriate resources for execution. By enabling cross-RSU task migration, idle RSUs with abundant terrestrial-level resources can be fully exploited, achieving system-wide workload balancing throughout the network. In addition, we enhance the predictive capabilities of RSU agents for future vehicular movements, to facilitate anticipatory direction determination of task migration.

Overall, through coordinated optimization and knowl-

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TABLE 2
Components of Decentralized Partially-Observable Markov Decision Process for Each Tier.

| Tiers of DRL Agents | Observations | Shared Observations | Actions | Rewards |
|---|---|---|---|---|
| UAV/Aerial-Tier: position scheduling | status of UAV $E_k^A$; statuses and workloads of RSUs within its flight range | status of UAV $E_k^A$; statuses and workloads of all RSUs in the system | which RSU in set $\mathcal{E}_k^R$ to hover over before offloading service delivery | negative cost of all tasks within its flight range: $-\sum_{E_i^R \in \mathcal{E}_k^R} \sum_{j=1}^{M_i^t} \text{cost}_{i,j}^t$ |
| RSU/Terrestrial-Tier: computation offloading & resource management | status of UAV $E_k^A$ that may cover the intersection; statuses of RSU $E_i^R$ and vehicle $v_{i,j}^t$; attributes of task $ta_{i,j}^t$ | status of UAV $E_k^A$ that may cover the intersection; statuses of all RSUs and vehicle $v_{i,j}^t$; attributes of task $ta_{i,j}^t$ | which device to execute the task; bandwidth to be allocated; priority score to be assigned | negative task completion cost: $-(w_l \cdot \text{latency} + w_e \cdot \text{energy} + \text{penalty})$ |

edge sharing, the two tiers of agents engage in mutual feedback and iterative policy refinement, effectively maintaining responsive and robust workload balancing across the network through seamless aerial-terrestrial cooperation.

### 4.2 Formalization of Dec-POMDP

Considering the constrained information perception of both UAV and RSU agents, we model the overall problem as a decentralized Partially-Observable Markov Decision Process (Dec-POMDP). Formally, it is defined as

$$\mathcal{G} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{O}_i\}_{i\in\mathcal{I}}, \{\mathcal{A}_i\}_{i\in\mathcal{I}}, \mathcal{P}, r, \gamma \rangle , \tag{27}$$

where $\mathcal{I}=\mathcal{I}^A \cup \mathcal{I}^R$ denotes the set of all agents, and $\mathcal{P}(s_{t+1} \mid s_t, a_t)$ denotes the shared state transition dynamics under a unified environment. The components are specified below.

***State Space.*** At time slot $t$, the global state representation $s^t \in \mathcal{S}$ describes the current condition of each component in the system. Specifically, it consists of information from all the UAVs, RSUs, vehicles, and generated computation tasks:

$$s^t = \big\{\{(f_k^A, n_k^A, q_{k,t}^A, B_k^A)\}_{k=1}^{N_A}, \{(f_i^R, n_i^R, q_{i,t}^R, B_i^R, e_i^R)\}_{i=1}^{N_R}, \{(u_{i,j}^t, p_{i,j}^t, e_{i,j}^t), (s_{i,j}^t, c_{i,j}^t, d_{i,j}^t)\}_{i=1,j=1}^{N_R, M_i^t}\big\}, \tag{28}$$

which is not fully observable for individual agents.

***Observation Space.*** Due to limited scope of information perception, each UAV agent can only obtain local information about itself and the RSUs within its flight range. We define $o_k^t = \big\{(f_k^A, n_k^A, q_{k,t}^A, B_k^A), \{(f_i^R, n_i^R, q_{i,t}^R, B_i^R, e_i^R), (M_i^t, S_i^t, C_i^t)\}_{E_i^R \in \mathcal{E}_k^R}\big\} \in \mathcal{O}_k^A$ as the partial observation of UAV agent $A_k^A$ at time slot $t$. Note that $M_i^t$, $S_i^t$, and $C_i^t$ represent the total number, data size, and required CPU cycles of all tasks covered by RSU $E_i^R$, which explicitly reflect its total workload at the current time. Based on the resource availability and workload conditions of all candidate RSUs in $\mathcal{E}_k^R$, UAV agent $A_k^A$ will select one to hover over, providing supplementary computing services. Similarly, each RSU agent is restricted to observing the status of itself, its covered vehicles, and the UAV that may assist it. Since computation offloading and resource management decisions are made on a per-vehicle basis, the partial observation of RSU agent $A_i^R$ for vehicle $v_{i,j}^t$ is $o_{i,j}^t = \big\{(f_k^A, n_k^A, q_{k,t}^A, B_k^A), (f_i^R, n_i^R, q_{i,t}^R, B_i^R, e_i^R), (u_{i,j}^t, p_{i,j}^t, e_{i,j}^t), (s_{i,j}^t, c_{i,j}^t, d_{i,j}^t)\big\} \in \mathcal{O}_i^R$, where $E_i^R \in \mathcal{E}_k^R$.

***Shared Observation Space.*** Although agents generate actions in a decentralized manner, they must cooperate to achieve the overall system objective; otherwise, they may easily fall into local optima. Therefore, we construct shared observations for agents by embedding specific global features into their original partial observations, to facilitate the two-tiered centralized training of agents. The shared observation of UAV agent $A_k^A$ includes the current conditions of all RSUs, i.e., $\tilde{o}_k^t = \big\{(f_k^A, n_k^A, q_{k,t}^A, B_k^A), \{(f_i^R, n_i^R, q_{i,t}^R, B_i^R, e_i^R), (M_i^t, S_i^t, C_i^t)\}_{i=1}^{N_R}\big\} \in \tilde{\mathcal{O}}_k^A$. By perceiving overall terrestrial conditions during training, UAVs are capable of identifying large-scale load aggregations spanning multiple flight ranges and developing aerial-level strategies for collaborative position scheduling. As for RSU agent $A_i^R$, its shared observation is denoted as $\tilde{o}_{i,j}^t = \{o_{i,j}^t, \{(f_{i'}^R, n_{i'}^R, q_{i',t}^R, B_{i'}^R, e_{i'}^R)\}_{i'=1, i'\neq i}^{N_R}\} \in \tilde{\mathcal{O}}_i^R$. The additional features provide valuable insights into the occupancy and idleness of different RSUs, supporting cross-RSU task migration and terrestrial-level collaboration.

***Action Space.*** The two tiers of agents make decentralized decisions in sequence. Based on the observation $o_k^t$, each UAV agent selects an intersection from $\mathcal{E}_k^R$ to hover over and establish communication with. We define the position scheduling action of UAV agent $A_k^A$ at time slot $t$ as $a_k^{s,t} \in \mathcal{A}_k^A$. Once UAVs are deployed to their designated positions, RSU agents can proceed to make further decisions for their covered vehicles. Given the UAV deployment status and the partial observation $o_{i,j}^t$, each RSU agent determines an appropriate device for executing task $ta_{i,j}^t$, along with the allocated bandwidth ratio and assigned priority score. The corresponding action of RSU agent $A_i^R$ for its covered vehicle $v_{i,j}^t$ is denoted as $a_{i,j}^t = \{a_{i,j}^{o,t}, a_{i,j}^{b,t}, a_{i,j}^{p,t}\} \in \mathcal{A}_i^R$. Note that the computation offloading decision $a_{i,j}^{o,t}$ is discrete, whereas the resource management decisions $a_{i,j}^{b,t}$ and $a_{i,j}^{p,t}$ are continuous, forming a hybrid action space that requires specialized consideration and handling.

***Reward.*** With the objective of minimizing both task completion latency and energy consumption, we define the immediate reward of each task as $r_{i,j}^t = -(w_l l_{i,j}^t + w_e e_{i,j}^t + \rho_{i,j}^t)$, where a penalty term $\rho_{i,j}^t = max(l_{i,j}^t - d_{i,j}^t, 0)$ is incorporated as reward shaping to penalize tasks failing to meet their deadlines. Accordingly, the system-level immediate reward at time slot $t$ is defined as $R_t = \sum_{i=1}^{N_R} \sum_{j=1}^{M_i^t} r_{i,j}^t$. The original minimization objective in (19) can be recast as maximizing the expected discounted cumulative return:

$$\max_{\Theta} J(\Theta) = \mathbb{E}_{\tau\sim(\pi_\Theta, \mathcal{P})} \left[ \sum_{t=1}^{T} \gamma^{t-1} R_t \right], \tag{29}$$

where $\Theta$ denotes the parameters of the joint policy $\pi_\Theta$, and $\tau$ is a trajectory induced by $\pi_\Theta$ and the transition dynamics.

Exact planning in a Dec-POMDP requires reasoning over joint action-observation histories or belief states, which is intractable in our NP-hard, high-dimensional, and dynamic

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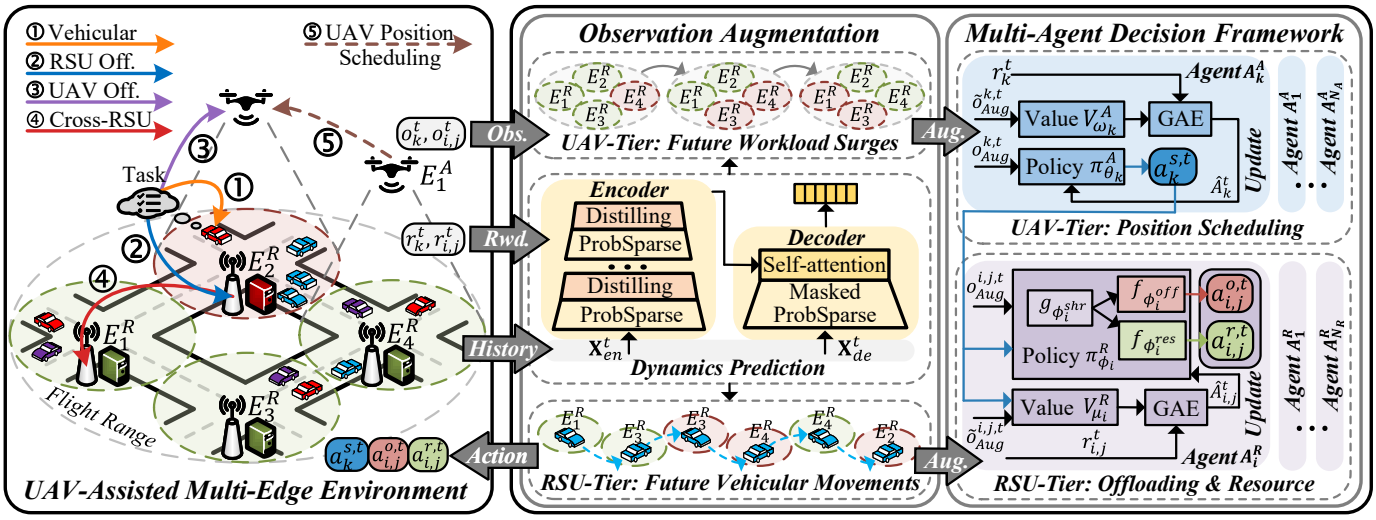



Fig. 4. Workflow of our Dynamics-Aware Hierarchical Multi-Agent (DHMA) deep reinforcement learning approach.

setting. We therefore adopt DRL as a scalable approximate solver to learn parameterized policies that map local observations to actions. Motivated by the intrinsic sequential dependency and distinct decision scopes of the two tiers, we adopt the following conditional factorization of the joint policy, yielding two sequentially coupled decision stages:

$$\pi_\Theta(a_t \,|\, o_t) = \prod_{k=1}^{N_A} \pi^A_{\theta_k}(a_k^{s,t} \,|\, o_k^t) \prod_{i=1}^{N_R} \pi^R_{\phi_i}(a_{i,j}^t \,|\, o_{i,j}^t, a_k^{s,t}). \quad (30)$$

Unlike a flat factorization, this formulation conditions RSU-tier policies on the previously determined UAV-tier actions, explicitly encoding the placement-before-service decision order. The above system objective is further decomposed into tier-specific long-term discounted returns. Specifically, the return is $\mathcal{R}_i^R = \sum_{t=1}^{T} \sum_{j=1}^{M_i^t} \gamma^{t-1} r_{i,j}^t$ for RSU agent $A_i^R$ and $\mathcal{R}_k^A = \sum_{E_i^R \in \mathcal{E}_k^R} \mathcal{R}_i^R$ for UAV agent $A_k^A$, where $\gamma$ is the discount factor. These tier-specific returns provide localized training signals for multi-agent policy optimization while remaining aligned with the global system objective.

### 4.3 Two-Tiered Hierarchical Multi-Agent Decision Framework

To implement the conditional policy factorization in equation (30), we develop a two-tiered hierarchical multi-agent decision framework. At both tiers, each agent employs two neural networks: a policy network that generates actions from partial observations and a value network that evaluates the current policy using shared observations. This centralized training and decentralized execution paradigm enables cooperative yet localized decision-making within each tier, which is particularly critical for adapting to diverse traffic patterns across intersections and regions.

#### 4.3.1 Decentralized Action Generation

The action generation process of the two-tiered hierarchical multi-agent decision framework follows a sequential procedure, proceeding from the UAV tier to the RSU tier.

***UAV-Tier: Position Scheduling with Policy Network $\pi^A_{\theta_k}$.*** Given the partial observation $o_k^t$, the policy network $\pi^A_{\theta_k}$ of each UAV agent $A_k^A$ outputs a probability distribution $\pi^A_{\theta_k}(a_k^{s,t} | o_k^t)$ over the set of candidate RSUs within its flight range. A position scheduling action $a_k^{s,t}$ is then drawn from the distribution, indicating which RSU the UAV $E_k^A$ will hover over during time slot $t$. In order to effectively utilize the aerial-level resources, the deployment statuses of UAVs are forwarded into the RSU tier once all UAV positions have been determined. Thus, a new observation $o_{i,j}^{s,t} = \{o_{i,j}^t, a_k^{s,t}\}$ of vehicle $v_{i,j}^t$ is constructed for RSU $E_i^R$, by concatenating its original observation with the generated positioning decision of UAV $E_k^A$ that may cover the intersection.

***RSU-Tier: Computation Offloading & Resource Management with Policy Network $\pi^R_{\phi_i}$.*** Based on the new observations, each RSU agent $A_i^R$ is responsible for directing its covered tasks to suitable devices and assigning appropriate resources for execution. Given that the action space of the RSU tier involves both discrete and continuous variables, we design the policy network $\pi^R_{\phi_i}$ of each RSU agent $A_i^R$ to include both shared feature extraction layers $g(\cdot; \phi_i^{shr})$ and two task-specific heads: a discrete branch $f(\cdot; \phi_i^{off})$ for offloading decisions, and a continuous branch $f(\cdot; \phi_i^{res})$ for allocation and priority decisions. After obtaining the shared representation $h_{i,j}^t = g(o_{i,j}^{s,t}; \phi_i^{shr})$, actions are sampled from the corresponding probability distributions generated by the task-specific heads, incorporating specially designed masking operations as follows:

$$a_{i,j}^{o,t} \sim f(h_{i,j}^t, m_{i,j}^{d,t}; \phi_i^{off}),\ m_{i,j}^{d,t} = \mathcal{M}_d(a_k^{s,t}) \quad (31)$$

$$a_{i,j}^{r,t} = \{a_{i,j}^{b,t}, a_{i,j}^{p,t}\} \sim f(h_{i,j}^t; \phi_i^{res}),\ m_{i,j}^{c,t} = \mathcal{M}_c(a_{i,j}^{o,t}) \quad (32)$$

$$\{a_{i,j}^{b,t}, a_{i,j}^{p,t}\} \leftarrow \{a_{i,j}^{b,t}, a_{i,j}^{p,t}\} \cdot m_{i,j}^{c,t}. \quad (33)$$

Specifically, task $ta_{i,j}^t$ can only be offloaded to UAV $E_k^A$ if the UAV is deployed above the exact intersection where vehicle $v_{i,j}^t$ is located. Thus, a soft masking operation $\mathcal{M}_d$, which depends on the UAV deployment status, is employed during the discrete action sampling process to suppress the probabilities of infeasible UAV offloading actions towards zero for certain vehicles. Moreover, since at most one task can be assigned for local execution per time slot, a hard masking operation $\mathcal{M}_c$ is applied to the continuous actions, thereby eliminating the need for resource management in

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**Algorithm 1:** Sequential Procedure of Decentralized Action Generation during Data Collection at $t$.

**Input:** UAV policy network $\pi^A_{\theta^A_k}$, RSU policy network $\pi^R_{\phi_i}$, replay buffers $B^A_k$ and $B^R_i$, $\forall k, \forall i$.

**for** *each UAV agent* $A^A_k$, *k=1 to* $N_A$ **do**
  Obtain observation $o^t_k$ and shared observation $\tilde{o}^t_k$ from the environment;
  Draw a position scheduling action $a^{s,t}_k$ from the probability distribution $\pi^A_{\theta^A_k}(a^{s,t}_k|o^t_k)$;
**end**
**for** *each RSU agent* $A^R_i$, *i=1 to* $N_R$ **do**
  **for** *each vehicle* $v^t_{i,j}$, *j=1 to* $M^t_i$ **do**
    Obtain observation $o^{s,t}_{i,j} = \{o^t_{i,j}, a^{s,t}_k\}$ and shared observation $\tilde{o}^{s,t}_{i,j} = \{\tilde{o}^t_i, a^{s,t}_k\}$, where $E^R_i \in \mathcal{E}^R_k$;
    Get the shared representation $h^t_{i,j} = g(o^{s,t}_{i,j}; \phi^{shr}_i)$;
    Sample an offloading action $a^{o,t}_{i,j}$ from $f(\cdot;\phi^{off}_i)$ with a soft masking $\mathcal{M}_d$;
    Sample resource management actions $\{a^{b,t}_{i,j}, a^{p,t}_{i,j}\}$ from $f(\cdot;\phi^{res}_i)$ and apply a hard masking $\mathcal{M}_c$;
  **end**
**end**
Execute the joint action $a^t = \{\{a^{s,t}_k\}^{N_A}_{k=1}, \{a^t_{i,j}\}^{N_R, M^t_i}_{i=1,j=1}\}$ in the environment;
**for** *each UAV agent* $A^A_k$, *k=1 to* $N_A$ **do**
  Obtain reward $r^t_k$, next observation $o^{t+1}_k$, next shared observation $\tilde{o}^{t+1}_k$ from the environment;
  Store transition $(o^t_k, \tilde{o}^t_k, a^{s,t}_k, r^t_k, o^{t+1}_k, \tilde{o}^{t+1}_k)$ into $B^A_k$;
**end**
**for** *each RSU agent* $A^R_i$, *i=1 to* $N_R$ **do**
  Obtain reward $r^t_{i,j}$, next observation $o^{s,t'}_{i,j}$, next shared observation $\tilde{o}^{s,t'}_{i,j}$ of each vehicle $v^t_{i,j}$;
  Store transitions $\{(o^{s,t}_{i,j}, \tilde{o}^{s,t}_{i,j}, a^t_{i,j}, r^t_{i,j}, o^{s,t'}_{i,j}, \tilde{o}^{s,t'}_{i,j})\}^{M^t_i}_{j=1}$ of all covered vehicles into $B^R_i$;
**end**

this context. The whole process can be denoted as $a^t_{i,j} \sim \pi^R_{\phi_i}(a^t_{i,j}|o^{s,t}_{i,j})$ for simplicity.

***Overall:*** After performing the joint action of both tiers, $a^t = \{\{a^{s,t}_k\}^{N_A}_{k=1}, \{a^t_{i,j}\}^{N_R, M^t_i}_{i=1,j=1}\}$, the environment transitions to the next state and provides immediate feedback to all agents with respective rewards. Notably, the action generation process described above differs between the model training and execution phases. During data collection in the training phase, actions are stochastically sampled from the policy distributions to promote sufficient exploration. The resulting experience tuples are stored separately in the corresponding replay buffers of UAV and RSU agents for subsequent model updates, as illustrated in Algorithm 1. By contrast, during decision-making in the execution phase, the most probable (or mean) actions are deterministically selected from the learned policy distributions to ensure stable and reliable performance.

### 4.3.2 Centralized Model Update

To enable collaboration among agents, shared information is gathered and leveraged to facilitate the centralized model updates at each tier. Specifically, the value network in each agent is allowed to obtain additional global features that are unavailable to the policy network, providing richer supervisory signals for effective policy evaluation and optimization. Such access to system-wide knowledge helps mitigate the risk of convergence to local optima by encouraging globally informed decision-making. The updates are performed independently for each tier based on the experience tuples collected over $T$ time slots within each episode.

***UAV-Tier: Policy Optimization with Value Network $V^A_{\omega_k}$.*** Denoting the value network of UAV agent $A^A_k$ as $V^A_{\omega_k}$, the advantage value $\hat{A}^t_k$ is estimated using truncated Generalized Advantage Estimation during sample generation:

$$\hat{A}^t_k = \delta^t_k + (\gamma\lambda)\delta^{t+1}_k + \cdots + (\gamma\lambda)^{T-t}\delta^T_k, \tag{34}$$

$$\delta^t_k = r^t_k + \gamma V^A_{\omega^{old}_k}(\tilde{o}^{t+1}_k) - V^A_{\omega^{old}_k}(\tilde{o}^t_k), \tag{35}$$

where $\lambda$ is the GAE hyper-parameter. We measure the magnitude of policy update by the probability ratio between the current and previous policies, and apply a clipping operation with a hyper-parameter $\epsilon$ to ensure stability:

$$r_t(\theta_k) = \pi^A_{\theta_k}(a^{s,t}_k|o^t_k)/\pi^A_{\theta^{old}_k}(a^{s,t}_k|o^t_k), \tag{36}$$

$$r^{clip}_t(\theta_k) = clip(r_t(\theta_k), 1-\epsilon, 1+\epsilon). \tag{37}$$

The optimization objective of the policy network $\pi^A_{\theta_k}$ is defined as to maximize the following function:

$$\hat{\mathcal{L}}(\theta_k) = \hat{\mathbb{E}}_t[\min(r_t(\theta_k)\hat{A}^t_k, r^{clip}_t(\theta_k)\hat{A}^t_k) + c\mathcal{H}[\pi^A_{\theta_k}](o^t_k)]. \tag{38}$$

The first term denotes the clipped surrogate objective, which restricts the extent of policy update to stabilize the training process. The second term is an entropy bonus weighted by $c$, which is introduced to promote exploration during training.

The goal of updating the value network $V^A_{\omega_k}$ is to minimize the discrepancy between the predicted state values and the actual returns. Thus, the loss function can be defined using the clipped Mean Squared Error (MSE) function:

$$\hat{\mathcal{L}}(\omega_k) = \frac{1}{2}\hat{\mathbb{E}}_t[\max((V^A_{tgt} - V^A_{\omega_k})^2, (V^A_{tgt} - V^{A,clip}_{\omega_k})^2)], \tag{39}$$

$$V^A_{tgt}(\tilde{o}^t_k) = \hat{A}^t_k + V^A_{\omega^{old}_k}(\tilde{o}^t_k), \tag{40}$$

$$V^{A,clip}_{\omega_k} = V^A_{\omega^{old}_k} + clip(V^A_{\omega_k} - V^A_{\omega^{old}_k}, -\epsilon, +\epsilon). \tag{41}$$

The actual return $V^A_{tgt}(\tilde{o}^t_k)$ is calculated as the sum of the advantage estimate and predicted state value $V^A_{\omega^{old}_k}(\tilde{o}^t_k)$ before update. To avoid drastic changes that could destabilize the learning process, clipping is employed to ensure that the current prediction $V^A_{\omega_k}(\tilde{o}^t_k)$ does not deviate too much from the old one, which is controlled by the hyper-parameter $\epsilon$.

Note that each value network $V^A_{\omega_k}$ takes its shared observation $\tilde{o}^t_k$ as input, estimating the state values based on the current statuses and workloads of all RSUs. This holistic assessment of the current state allows for a more comprehensive understanding of terrestrial conditions across different UAV flight ranges, guiding the policy network $\pi^A_{\theta_k}$ to update towards coordinated behavior that benefits the entire group. Thereby, when confronted with large-scale load aggregations spanning multiple flight ranges, UAVs can collaboratively perceive the traffic conditions and dynamically schedule themselves to deliver on-demand aerial capacity to overloaded RSUs responsively.

***RSU-Tier: Policy Optimization with Value Network $V^R_{\mu_i}$.*** The policy and value networks of each RSU agent $A^R_i$ are updated following a process similar to the UAV tier, with one critical distinction: the replay buffer of each

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RSU agent accumulates $\Gamma = \sum_{t=1}^{T} M_i^t$ experience tuples due to the necessity of per-vehicle computation offloading and resource management. By re-indexing these experience tuples using $\tau$ that ranges from 1 to $\Gamma$, we establish the bijective mapping $o_i^\tau \equiv o_{i,j}^{s,t}$, which extends consistently to all other relevant notations at the RSU tier.

Therefore, the objective function for updating the policy network $\pi_{\phi_i}^R$ can be defined as follows, with the goal of maximizing the estimated advantages while maintaining the training stability:

$$\hat{\mathcal{L}}(\phi_i) = \hat{\mathbb{E}}_\tau[\min(r_\tau(\phi_i)\hat{A}_i^\tau, r_\tau^{clip}(\phi_i)\hat{A}_i^\tau) + c\mathcal{H}[\pi_{\phi_i}^R](o_i^\tau)]. \quad (42)$$

The advantage estimate $\hat{A}_i^\tau$ quantifies the excess value of a specific action over the expected return under current policy. In addition, an entropy regularization term $c\mathcal{H}[\pi_{\phi_i}^R](o_i^\tau)$ is introduced to balance exploration and exploitation.

With the shared observation $\tilde{o}_i^\tau$ as input, each RSU agent $A_i^R$ optimizes its value network $V_{\mu_i}^R$ to estimate the state values as accurately as possible relative to the actual returns. This is achieved by minimizing the following training loss:

$$\hat{\mathcal{L}}(\mu_i) = \frac{1}{2}\hat{\mathbb{E}}_\tau[\max((V_{tgt}^R(\tilde{o}_i^\tau) - V_{\mu_i}^R(\tilde{o}_i^\tau))^2, (V_{tgt}^R(\tilde{o}_i^\tau) - V_{\mu_i}^{R,clip}(\tilde{o}_i^\tau))^2)]. \quad (43)$$

Notably, by incorporating real-time statuses gathered from all RSUs, the value network $V_{\mu_i}^R$ performs policy evaluation based on a comprehensive system-wide assessment of RSU occupancy and idleness. This global perspective on resource availability encourages the policy network $\pi_{\phi_i}^R$ to optimize for terrestrial-level collaboration, directing tasks towards appropriate devices during cross-RSU migration. Consequently, idle RSUs with abundant resources can be fully exploited to dynamically absorb excessive tasks from overloaded ones, establishing system-wide workload balancing among coordinated RSUs throughout the network.

**Algorithm 2:** Training Procedure of the Two-Tiered Hierarchical Multi-Agent Decision Framework.

Initialize UAV policy $\pi_{\theta_k}^A$ and value $V_{\omega_k}^A$ networks, initialize $\pi_{\theta_k^{old}}^A$ with $\theta_k^{old} \leftarrow \theta_k$, $V_{\omega_k^{old}}^A$ with $\omega_k^{old} \leftarrow \omega_k$, $\forall k$;
Initialize RSU policy $\pi_{\phi_i}^R$ and value $V_{\mu_i}^R$ networks, initialize $\pi_{\phi_i^{old}}^R$ with $\phi_i^{old} \leftarrow \phi_i$, $V_{\mu_i^{old}}^R$ with $\mu_i^{old} \leftarrow \mu_i$, $\forall i$;
Initialize replay buffers $B_k^A$ and $B_i^R$, $\forall k$, $\forall i$.
**for** *episode = 1 to episode-length* **do**
  Reset the environment and agents' statuses;
  /* Decentralized Action Generation */
  **for** *time slot t=1 to T* **do**
    Collect experience tuples into the replay buffers with $\pi_{\theta_k^{old}}^A$ and $\pi_{\phi_i^{old}}^R$ via Algorithm 1 for $\forall k$, $\forall i$;
    Re-index tuples in buffer $B_i^R$ for each RSU agent $A_i^R$ according to bijective mapping $o_i^\tau \equiv o_{i,j}^{s,t}$;
    Estimate advantage values $\hat{A}_k^t$ and $\hat{A}_i^\tau$ by GAE;
    Compute actual returns $V_{tgt}^A(\tilde{o}_k^t)$ and $V_{tgt}^R(\tilde{o}_i^\tau)$;
  **end**
  /* Centralized Model Update */
  **for** *each UAV agent $A_k^A$, k=1 to $N_A$* **do**
    **for** *mini-batch $\mathcal{M}_k^A$ in replay buffer $B_k^A$* **do**
      Update policy network $\pi_{\theta_k}^A$ by maximizing $\hat{\mathcal{L}}(\theta_k)$ as defined in equation (38);
      Update value network $V_{\omega_k}^A$ by minimizing $\hat{\mathcal{L}}(\omega_k)$ as defined in equation (39);
    **end**
    Clear replay buffer $B_k^A$;
    Update $\theta_k^{old} \leftarrow \theta_k$ and $\omega_k^{old} \leftarrow \omega_k$;
  **end**
  **for** *each RSU agent $A_i^R$, i=1 to $N_R$* **do**
    **for** *mini-batch $\mathcal{M}_i^R$ in replay buffer $B_i^R$* **do**
      Update policy network $\pi_{\phi_i}^R$ by maximizing $\hat{\mathcal{L}}(\phi_i)$ as defined in equation (42);
      Update value network $V_{\mu_i}^R$ by minimizing $\hat{\mathcal{L}}(\mu_i)$ as defined in equation (43);
    **end**
    Clear replay buffer $B_i^R$;
    Update $\phi_i^{old} \leftarrow \phi_i$ and $\mu_i^{old} \leftarrow \mu_i$;
  **end**
**end**

### 4.4 Dynamics-Aware Multi-Agent Observation Augmentation

While cooperative training is enabled by value networks with specially designed shared observations, we argue that decision-making based solely on reactive observations remains inadequate in highly dynamic vehicular environments. Such reliance on only instantaneous observations is prone to hysteresis, where delayed responses to dynamic traffic changes—such as bursty congestion or vehicle mobility shifts—can lead to suboptimal coordination and degraded system performance. Instead of passively waiting for observable changes, proactively predicting future dynamics, particularly RSU workload surges and vehicular movements, can significantly enhance the timeliness and effectiveness of collaboration among agents.

#### 4.4.1 Trajectory Prediction-Based Dynamics Perception

To enable such proactive perception of future dynamics, both RSU workload surges and vehicular movements are inferred through fine-grained vehicle trajectory prediction, as dynamic traffic changes are inherently driven by the high-speed and often irregular behavior of individual vehicles. This trajectory-level prediction not only captures latent mobility patterns, but also facilitates early identification of emerging traffic trends that may be overlooked by coarse-grained or aggregated features.

Given a historical sequence of fine-grained locations $\{\mathcal{X}_{i,j}^{t-l}\}_{l=0}^{H-1} = \{(x_{i,j}^{t-l}, y_{i,j}^{t-l})\}_{l=0}^{H-1}$, our goal is to predict the future trajectory sequence of vehicle $v_{i,j}^t$ over the next $B$ timesteps, as $\{\hat{\mathcal{X}}_{i,j}^{t+l}\}_{l=1}^{B} = \mathcal{F}_{TS}(\{\mathcal{X}_{i,j}^{t-l}\}_{l=0}^{H-1}; \Theta_{TS})$. The time-series forecasting module $\mathcal{F}_{TS}$ parameterized by $\Theta_{TS}$ is pre-trained prior to its integration with the two-tiered hierarchical multi-agent decision framework. Benefiting from the efficiency of Informer [47], we implement $\mathcal{F}_{TS}$ based on its encoder-decoder architecture.

In order to capture both local and global temporal contexts, a uniform input representation is initially employed:

$$\mathcal{X}_{\text{emb}}^{t-l} = w\mathbf{u}_{i,j}^{t-l} + \text{PE}_{(pos,)} + \sum\nolimits_p [\text{SE}_{(pos)}]_p, \quad (44)$$

where $\mathbf{u}_{i,j}^{t-l}$ denotes the $d$-dim projection of $\mathcal{X}_{i,j}^{t-l}$. $\text{PE}_{(pos,)}$ and $[\text{SE}_{(pos)}]_p$ represent the local positional embedding and the global learnable embedding for each hierarchical

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timestamp $p$, respectively. Thus, the historical sequence $\{\mathcal{X}_{i,j}^{t-l}\}_{l=0}^{H-1}$ can be shaped into a matrix $\mathbf{X}_{en}^{t} \in \mathbb{R}^{H\times d}$.

To achieve efficient computation and memory usage, the encoder employs multiple layers of ProbSparse self-attention and distilling to generate the hidden representations. Given matrices $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ derived from $\mathbf{X}_{en}^{t}$, the ProbSparse self-attention can be defined as $\mathcal{A}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{Softmax}(\overline{\mathbf{Q}}\mathbf{K}^{\top}/\sqrt{d})\mathbf{V}$, where $\overline{\mathbf{Q}}$ is a sparse matrix constructed by selecting the top-$u$ queries based on a sparsity criterion of Kullback-Leibler divergence. We denote the output of the $k$-th multi-head ProbSparse self-attention layer as $[\mathbf{X}_k^t]_{\mathcal{A}}$. To compress intermediate representations across layers, a distilling operation is applied, where $[\mathbf{X}_k^t]_{\mathcal{A}}$ is downsampled into its half slice via max-pooling, forming the input $\mathbf{X}_{k+1}^t$ for the next self-attention layer.

The decoder adopts a generative inference mechanism that predicts the entire future sequence in a single forward pass, thereby avoiding the inefficiency of step-by-step decoding. Specifically, we segment a start token $\{\mathcal{X}_{i,j}^{t-l}\}_{l=0}^{D-1}$ from $\{\mathcal{X}_{i,j}^{t-l}\}_{l=0}^{H-1}$ and append a zero-padded sequence of length $B$ as placeholders for future predictions. This resulting sequence is then embedded as a matrix $\mathbf{X}_{de}^{t} \in \mathbb{R}^{(D+B)\times d}$ and forwarded into the decoder, which comprises a masked multi-head ProbSparse self-attention layer followed by a standard multi-head self-attention layer. The predicted trajectory sequence $\{\hat{\mathcal{X}}_{i,j}^{t+l}\}_{l=1}^{B}$ can be obtained through a fully connected output layer in the final step.

The fine-grained prediction of vehicle trajectories provides valuable insights into the future behavior of individual vehicles within the system. By further transforming the predicted raw traces into structured knowledge of RSU workload distributions and vehicle mobility patterns, agents are empowered to perceive future dynamics proactively.

#### 4.4.2 Two-Tiered Multi-Agent Observation Augmentation

To facilitate timely and effective multi-agent collaboration, predicted trajectories are reorganized and incorporated into the observations of agents at both tiers, offering foresight-driven guidance for their respective decision-making.

***UAV-Tier: Observation Augmentation with Future RSU Workload Surges.*** Triggered by unexpected events such as sudden traffic accidents, temporary road closures, or large-scale gatherings near intersections, workload surges frequently occur due to the rapid aggregation of computation tasks at certain RSUs, resulting in time-varying hot zones characterized by localized computational bottlenecks. UAVs are deployed as aerial-level servers with the purpose of providing responsive offloading support by dynamically scheduling themselves towards overloaded RSUs. However, without predictive awareness of such surges, UAV agents can only respond reactively after RSU performance degradation becomes observable, leading to delayed intervention and inefficient resource utilization. Therefore, to ensure preemptive position scheduling of UAVs, we strengthen their perception capabilities for future workload surges.

Specifically, we leverage the predicted vehicle trajectories to infer future spatio-temporal workload distributions. Based on the fine-grained location predictions $\{\hat{\mathcal{X}}_{i,j}^{t+l}\}_{l=1}^{B}$, each vehicle $v_{i,j}^{t}$ is mapped to its nearest RSU at every future timestep, yielding an RSU association sequence $\{\hat{\mathcal{Y}}_{i,j}^{t+l}\}_{l=1}^{B}$ along its predicted path. By aggregating vehicle counts associated with each RSU over the prediction horizon, we construct a future workload distribution profile $\{\{\hat{\mathcal{Z}}_i^{t+l}\}_{l=1}^{B}\}_{i=1}^{N_R}$ over all RSUs. This profile is then embedded into the observation and shared observation of each UAV agent as

$$o_{Aug}^{k,t} = \{o_k^t, \{\{\hat{\mathcal{Z}}_i^{t+l}\}_{l=1}^{B}\}_{E_i^R\in\mathcal{E}_k^R}\}, \tag{45}$$

$$\tilde{o}_{Aug}^{k,t} = \{\tilde{o}_k^t, \{\{\hat{\mathcal{Z}}_i^{t+l}\}_{l=1}^{B}\}_{i=1}^{N_R}\}. \tag{46}$$

This augmentation enables UAV agents to preemptively identify and reposition towards potential hot zones before overloads materialize. By incorporating global workload trends into shared observations, UAV agents can coordinate their actions based on a collective understanding of the evolving task demands, thereby delivering aerial-level computing services in a timely and cooperative manner.

***RSU-Tier: Observation Augmentation with Future Vehicular Movements.*** Beyond workload surges, spatio-temporal load imbalances commonly arise across intersections, driven by regional functionality, road network topology, and human activity, leaving some RSUs saturated while others underutilized. Cross-RSU task migration serves as a lightweight and cost-effective mechanism to achieve system-wide workload balancing through dynamic task redistribution. However, its effectiveness critically depends on the alignment with the moving directions of vehicles. Without foresight into future vehicular movements, RSU agents are constrained to making myopic decisions based solely on current resource occupancy, potentially misdirecting tasks towards underutilized RSUs that vehicles are moving away from. Such misalignment introduces unnecessary communication overhead and compromises the intended benefits of workload balancing. Therefore, to enable anticipatory cross-RSU task migration, we enhance RSU agents with predictive perception of future vehicular movements.

In particular, the predicted vehicle trajectories are assigned to the corresponding RSU agents to facilitate the identification of future mobility patterns for their covered vehicles. Rather than directly using the predicted raw trajectory, we enrich both the partial and shared observations of RSU agent $A_i^R$ for vehicle $v_{i,j}^t$ with the previously obtained RSU association sequence $\{\hat{\mathcal{Y}}_{i,j}^{t+l}\}_{l=1}^{B}$, which explicitly captures the specific RSUs that vehicle $v_{i,j}^t$ is expected to traverse over time. Since multi-hop wireless transmission is adopted during task migration, this also enables RSU agents to assess migration overhead through the number of hops between the current and target RSUs. Consequently, the augmented representations are constructed as

$$o_{Aug}^{i,j,t} = \{o_{i,j}^t, a_k^{s,t}, \{\hat{\mathcal{Y}}_{i,j}^{t+l}\}_{l=1}^{B}\}, \tag{47}$$

$$\tilde{o}_{Aug}^{i,j,t} = \{\tilde{o}_{i,j}^t, a_k^{s,t}, \{\hat{\mathcal{Y}}_{i,j}^{t+l}\}_{l=1}^{B}\}. \tag{48}$$

Such anticipatory augmentation empowers RSU agents to make strategic migration decisions that prioritize underutilized RSUs along vehicles' future routes, thereby improving the coordinated exploitation of terrestrial-level computing resources for system-wide workload balancing.

### 4.5 Complexity Analysis

The time complexity of our DHMA approach is given by $\mathcal{O}\big(H\log H + \big((4+(8+B)|\mathcal{E}_k^R|)h + h^2 + h|\mathcal{E}_k^R|\big) + \bar{M}\big((15+B+$

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$|\mathcal{E}_k^R|)h+h^2+h(N_R+4))) = \mathcal{O}(H\log H+\bar{M}(B+h)h)$, where $h$ is the hidden embedding size. Each RSU covers an average of $\bar{M}$ vehicles; since $\bar{M}$ is typically much smaller than $h$, the dominant term $\mathcal{O}(\bar{M}h^2)$ remains controlled, supporting scalability to large-scale deployments. The space complexity is $\mathcal{O}(H\log H+(N_A+N_R)CS) = \mathcal{O}(H\log H+N_RCS)$, where $C$ is the buffer capacity and $S$ is the experience tuple size. Note that the replay-memory term $\mathcal{O}(N_RCS)$ is decoupled from the vehicle count, which enables scaling to real-world networks with massive vehicles.

### 4.6 Convergence Analysis

Let $\theta_A = \{\theta_k\}_{k=1}^{N_A}$ and $\phi_R = \{\phi_i\}_{i=1}^{N_R}$ be the policy parameters for UAV and RSU tiers. Let $\omega_A = \{\omega_k\}_{k=1}^{N_A}$ and $\mu_R = \{\mu_i\}_{i=1}^{N_R}$ denote the value-network parameters.

***Theorem 2 (Convergence).*** *Suppose that Assumptions 1-2 in Appendix B hold, and the initial parameters lie in a sufficiently small neighborhood of a local stationary solution $(\theta_A^\star, \phi_R^\star)$, where $\phi_R^\star \in \arg\max_{\phi_R} J_c(\theta_A^\star, \phi_R)$ and $\theta_A^\star$ is a strict local maximizer of $\bar{J}_c(\theta_A)$. Then, almost surely, $\omega_k \to \omega_k^\star(\theta_A^\star, \phi_R^\star)$, $\mu_i \to \mu_i^\star(\theta_A^\star, \phi_R^\star)$, $\phi_R \to \phi_R^\star$, and $\theta_A \to \theta_A^\star$.*

*Proof.* Standard ODE/stochastic-approximation arguments ensure almost-sure convergence. The detailed assumptions, auxiliary lemmas, and full proof are given in Appendix B.

## 5 Performance Evaluation

### 5.1 Evaluation Setup

#### 5.1.1 Simulation Scenarios

Given the practical challenges of limited accessibility and reproducibility associated with deploying and testing in large-scale real-world traffic environments, we utilize the widely adopted open-source traffic simulator SUMO [48], as in existing representative studies [1], [7], [13], [49]–[51], to generate reproducible road networks and vehicular traces for rigorous and controlled evaluation. Four simulation scenarios are constructed with diverse topological structures and traffic densities: two grid-based synthetic networks **Grid$_{3\times3}$** and **Net$_4$**, and two real-world networks **Pasubio** and **A.Costa** [52], as shown in Figure 5. The statistics for these scenarios are summarized in Table 3.

***Synthetic Networks: Grid$_{3\times3}$*** and ***Net$_4$*** are designed to evaluate general performance in typical grid-based scenarios. As shown in Figure 5a, Grid$_{3\times3}$ represents a standard 3×3 network, where each intersection is equipped with an RSU that provides computing services. In addition, 3 UAVs are employed as aerial-level servers, delivering supplementary resources to different regions. To further investigate performance under load imbalances, **Net$_4$** is adapted from a standard 4×4 network through selective road removal and lane reallocation. Besides, 4 UAVs are deployed with various flight ranges as depicted in Figure 5b.

TABLE 3
Statistics of simulation scenarios.

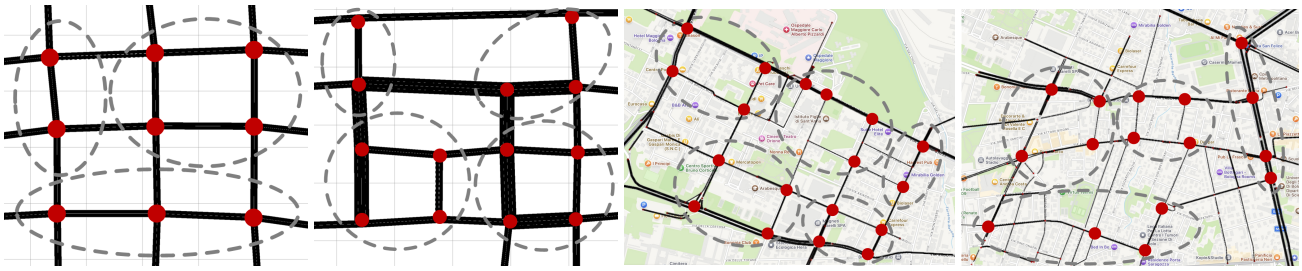
| Scenarios | Grid$_{3\times3}$ | Net$_4$ | Pasubio | A.Costa |
|---|---|---|---|---|
| Duration:Episodes | 10h:599 | 24h:1439 | 24h:1439 | 24h:1439 |
| #Vehicles | 12,000 | 32,400 | 30,955 | 27,051 |
| #UAVs / #RSUs | 3 / 9 | 4 / 13 | 5 / 18 | 4 / 16 |
| $f_k^A$ / $f_i^R$ (GHz) | 6 / 12 | 4 / 10 | 8 / 15 | 8 / 15 |
| #RSUs in UAV Ranges | 2, 4, 3 | 2, 3, 4, 4 | 4, 4, 2, 4, 4 | 4, 4, 4, 4 |
| Max. Flow/se | 19 ve/s | 24 ve/s | 64 ve/s | 28 ve/s |
| Avg. Flow/se | 3.6 ve/s | 6.8 ve/s | 13.4 ve/s | 5.9 ve/s |

***Real-World Networks: Pasubio*** and ***A.Costa*** are constructed based on real-world traffic regions in Bologna, Italy [52]. As shown in Figure 5c, Pasubio is located near a hospital and includes common routes to a football stadium. We implement RSUs at 18 major intersections and deploy 5 UAVs to provide additional coverage. A.Costa represents another area around the football stadium, where 16 intersections are configured with RSUs along with 4 UAVs flying above, as depicted in Figure 5d.

***Traffic Simulation:*** Vehicular traces of 10 and 24 hours are randomly generated for the two synthetic scenarios, where traffic volume is designed to remain relatively stable in Grid$_{3\times3}$ scenario, while experiencing a low-high-medium transition in Net$_4$ scenario. For the Pasubio and A.Costa scenarios, to closely approximate real-world traffic conditions, we generate 24-hour vehicular traces following the practical traffic patterns derived from real-world analysis [52], as shown in Figure 6. The vehicular traces are divided into one-minute episodes and there are $T = 60$ time slots of equal length $\Delta = 1$ s in each decision episode.

#### 5.1.2 Experimental Settings

The computing powers $f_k^A$ and $f_i^R$ of UAVs and RSUs are set as specified in Table 3. The channel bandwidths are set to $B_k^A = 8$ MHz and $B_i^R = 10$ MHz, respectively. The data transmission rate $\beta$ between adjacent RSUs is set to 500 Mbits/s or 1 Gbits/s. Each vehicle, with computing power $u_{i,j}^t = 2$ GHz, generates one computation task per second. Each task is randomly sampled with a data size $s_{i,j}^t$ in $[600, 900]$ Mbits, a required computation $c_{i,j}^t$ in $[3, 5]$ G cycles, and a deadline $d_{i,j}^t$ in $[3, 5]$ s. The transmit powers $e_{i,j}^t$ and $e_i^R$ are 200 mW and 250 mW, and the effective switched capacitance coefficient $\kappa$ is set to $10^{-28}$ as in [14]. The default



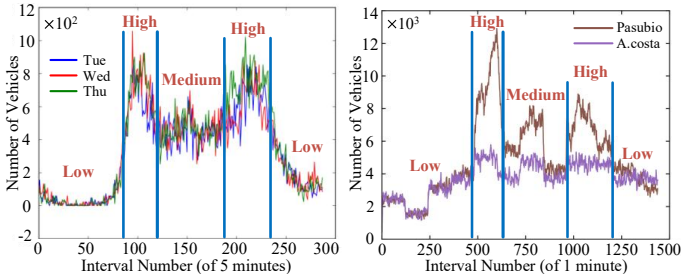
(a) Grid$_{3\times3}$ (b) Net$_4$ (c) Pasubio (d) A.Costa

Fig. 5. Topologies of four road networks. Red dots for RSUs and dashed circles for UAV flight ranges.

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(a) Real-world Measurements (b) Simulated with SUMO

Fig. 6. Region-level traffic volume of 24 hours in Pasubio and A.Costa. Our simulated data closely follows the patterns observed in real-world measurements from 636 detectors over three weekdays.

balance coefficients are set to $w_l = w_e = 0.5$, corresponding to a neutral setting that assigns equal importance to latency and energy. The coefficient pair can also be flexibly adjusted to suit different performance preferences. An additional sensitivity analysis is provided in Appendix D.1.

We set the size of the replay buffers to 30,000 and the discount factor $\gamma$ to 0.99. The Adam optimizer is employed with a linear schedule and an initial learning rate of 15e-5. During model update, each agent is trained for 15 epochs with 5 mini-batches. For the pre-training of $\mathcal{F}_{TS}$, we set $H = 32$, $B = 8$, $D = 16$. There are 2 encoder layers and 1 decoder layer with $d = 512$, and an 8-head ProbSparse self-attention is employed. Besides, we adopt the MSE loss and Adam optimizer with a learning rate of 1e-4.

#### 5.1.3 Baselines

To validate the effectiveness of our DHMA approach, we implement the following baselines for comparison.

***Heuristic Strategies*** serve as rule-based benchmarks that do not rely on data-driven optimization. As adopted in prior studies [2], [15], [53]–[55], they are adapted to fit our setting.

- *Vehicular Execution (**Veh-E**)*: vehicles process all generated tasks locally with their computing resources onboard.
- *RSU Offloading (**RSU-O**)*: vehicles offload all their tasks to the RSUs covering the corresponding intersections.
- *Proportion-$x$ Offloading (**PO-$x$%**)*: vehicles offload a fixed proportion $x\% \in \{20\%, 50\%, 80\%\}$ of tasks to their connected RSUs while handling the remaining tasks locally.
- *Stochastic Execution (**Sto-E**)*: each task has an equal probability of being executed locally, offloaded to its nearest RSU, offloaded to a UAV provided it is hovering overhead, or migrated to any other RSU in the system.
- *UAV-Priority (**UAV-P**)*: vehicles preferentially offload their tasks to UAVs when within coverage; if not, tasks are randomly handled via ground-based options.
- *Greedy Strategy (**Greedy**)*: UAVs are deployed above the RSUs with the highest workloads, while vehicles make greedy offloading decisions based on immediate rewards.

***DRL-Based Methods***, tailored for UAV position scheduling and vehicular computation offloading, are introduced as advanced learning-based approaches.

- ***M-DRL*** [31]: a single-agent framework that combines DQN and DDPG to enable task offloading and discretized resource allocation with LSTM for mobility prediction.
- ***TMP-DDPG*** [11]: a decentralized DDPG scheme with Transformer-based MD distribution prediction for joint computation offloading, task migration, and resource allocation in smart city networks.
- ***DE-TD3*** [34]: a bi-level optimization framework, where Differential Evolution is employed for multi-UAV deployment, and a single-agent TD3 algorithm is applied for edge-node task offloading and UAV resource allocation.
- ***AB-MAPPO*** [41]: an MAPPO approach with Beta distribution and attention mechanism for energy-efficient computation offloading in aerial edge networks. Two types of agents—MUs and UAVs—are involved, where MU agents are adapted to RSU agents to align with our work.
- ***MATD3*** [40]: an MATD3 framework where each UAV agent adopts a TD3 algorithm to simultaneously determine UAV movements, user task offloading, and server resource allocation in one combined policy.

The latter 3 baselines originally do not include dynamics prediction and are therefore evaluated without additional predictive inputs to preserve fidelity; see Appendix C (Table C.1) for a detailed comparison. For baselines with limited scalability, we cap the number of vehicles that can be handled and apply Sto-E to vehicles beyond the limit.

#### 5.1.4 Evaluation Metrics

Four key metrics are employed to provide a comprehensive analysis from multiple perspectives. The average results of all tasks generated during testing are reported.

- *Reward (**Rwd.**)*: refers to the immediate reward received for completing a task from the environment.
- *Task Completion Latency (**Lat.**)*: denotes the total time required to complete a task from initiation to final execution.
- *Task Energy Consumption (**En.**)*: denotes the total energy consumed during all processes of task completion.
- *Task Failure Rate (**FR**)*: is defined as the proportion of tasks that exceed their respective deadlines among all tasks. The number of failed tasks covered by RSU $E_i^R$ at $t$ is given by $\sum_{j=1}^{M_i^t} \mathbb{1}(l_{i,j}^t - d_{i,j}^t > 0)$, where $\mathbb{1}(\cdot)$ is an indicator function that yields 1 when its enclosed condition holds true.

### 5.2 Evaluation Results

#### 5.2.1 Convergence Analysis

As illustrated in Figure 7, our proposed DHMA approach demonstrates stable and efficient convergence behavior during the two-tiered centralized model training phase. Taking the $\text{Grid}_{3\times3}$ and $\text{Net}_4$ scenarios as representative examples, convergence rates exceeding 95% and 85% are respectively achieved after 100 episodes of parameter iteration. More specifically, the reward values of all UAV and RSU agents increase consistently until reaching stabilization, as shown in Figures 7a and 7c. A substantial decline in task failure rate is observed—dropping from nearly 80% in the early episodes to a stabilized 3% for $\text{Grid}_{3\times3}$ scenario, and from approximately 40% to just 2% upon convergence for $\text{Net}_4$ scenario. In addition, both task completion latency and energy consumption show significant reductions and eventually converge as training proceeds, which can be intuitively observed in Figures 7b and 7d. These results indicate that the two tiers of agents have effectively learned collaborative decision-making strategies through coordinated optimization and knowledge sharing, achieving system optimization objective through seamless aerial-terrestrial cooperation.

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TABLE 4
Performance comparison of our DHMA approach with different baselines on four simulation scenarios. The best results are **bolded**.

| Approaches | $\text{Grid}_{3\times3}$ | | | | $\text{Net}_4$ | | | | Pasubio | | | | A.Costa | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Rwd. | FR | Lat. | En. | Rwd. | FR | Lat. | En. | Rwd. | FR | Lat. | En. | Rwd. | FR | Lat. | En. |
| **Veh-E** | -43.925 | 0.9335 | 25.487 | 19.214 | -45.238 | 0.9370 | 26.188 | 19.745 | -49.184 | 0.9457 | 28.300 | 21.323 | -48.831 | 0.9446 | 28.111 | 21.182 |
| **RSU-O** | -46.350 | 0.7821 | 26.163 | 21.208 | -40.255 | 0.7261 | 22.980 | 18.268 | -86.753 | 0.6024 | 47.182 | 37.731 | -41.431 | 0.5820 | 23.153 | 19.096 |
| **PO-20%** | -21.838 | 0.7137 | 13.325 | 10.221 | -22.360 | 0.7190 | 13.621 | 10.385 | -24.944 | 0.7565 | 15.022 | 11.538 | -24.278 | 0.7302 | 14.624 | 11.214 |
| **PO-50%** | -7.9711 | 0.4548 | 5.6096 | 4.7861 | -6.8731 | 0.4019 | 4.9947 | 4.1291 | -18.686 | 0.4586 | 11.209 | 9.1953 | -8.1338 | 0.4159 | 5.5686 | 4.8113 |
| **PO-80%** | -24.364 | 0.5334 | 14.299 | 11.844 | -20.339 | 0.4760 | 12.167 | 9.8522 | -51.055 | 0.4256 | 28.162 | 22.739 | -22.272 | 0.3953 | 12.883 | 10.893 |
| **Sto-E** | -14.769 | 0.7900 | 8.4235 | 11.543 | -21.572 | 0.6544 | 12.419 | 12.828 | -19.808 | 0.6341 | 10.902 | 13.759 | -17.553 | 0.7753 | 9.0704 | 15.163 |
| **UAV-P** | -43.978 | 0.6959 | 24.245 | 22.369 | -54.883 | 0.6281 | 30.290 | 25.808 | -32.396 | 0.6156 | 17.790 | 18.179 | -28.560 | 0.7406 | 15.311 | 18.336 |
| **Greedy** | -7.2192 | 0.4147 | 5.2326 | 4.5376 | -5.1933 | 0.3692 | 4.2736 | 4.1803 | -15.337 | 0.4134 | 10.319 | 10.185 | -7.8435 | 0.3698 | 5.1747 | 5.4293 |
| **M-DRL** | -4.3836 | 0.3646 | 4.5281 | 4.2836 | -3.9273 | 0.2717 | 3.4298 | 3.2981 | -8.8467 | 0.3572 | 6.2319 | 6.0237 | -6.0127 | 0.2839 | 4.0239 | 4.1960 |
| **TMP-DDPG** | -4.1530 | 0.2938 | 3.7268 | 3.5837 | -3.7236 | 0.2389 | 2.9384 | 2.9198 | -7.9742 | 0.3287 | 5.3238 | 4.9109 | -5.3847 | 0.2743 | 3.7832 | 3.8103 |
| **DE-TD3** | -3.8123 | 0.2311 | 2.9531 | 2.8739 | -3.2273 | 0.1691 | 2.7561 | 2.6011 | -7.3831 | 0.2991 | 4.8928 | 4.5261 | -4.7233 | 0.2636 | 3.5873 | 3.4928 |
| **AB-MAPPO** | -2.2927 | 0.0872 | 2.1467 | 2.2972 | -2.7534 | 0.0795 | 2.4287 | 2.3182 | -4.6274 | 0.2418 | 3.4129 | 3.6987 | -3.4236 | 0.1927 | 2.8475 | 2.9321 |
| **MATD3** | -2.5861 | 0.1137 | 2.4412 | 2.5187 | -2.9436 | 0.0983 | 2.5374 | 2.5198 | -5.2478 | 0.2618 | 3.8273 | 3.7812 | -4.2310 | 0.2387 | 3.0219 | 3.1627 |
| **DHMA** | **-2.0145** | **0.0264** | **1.9244** | **2.0803** | **-2.3649** | **0.0661** | **2.1401** | **2.1820** | **-3.7985** | **0.2107** | **2.9317** | **3.1754** | **-2.8145** | **0.1469** | **2.4568** | **2.6710** |

### 5.2.2 *Performance Details*

Table 4 presents a comprehensive evaluation of our approach compared with established baselines. Across all four simulation scenarios, DHMA consistently achieves the highest rewards and lowest failure rates, while exhibiting superior performance in terms of both completion latency and energy consumption. These demonstrate the superiority and robustness of DHMA in achieving effective computation offloading through aerial-terrestrial cooperation.

In comparison, Veh-E, RSU-O, and UAV-P generally yield the poorest performance owing to their primary reliance on a single category of resources. Through basic vehicle-RSU cooperation, the PO-x% series attains a certain degree of improvements, with PO-50% offering the most balanced trade-off. Despite incorporating UAV assistance, Sto-E still exhibits persistently high failure rates (63%-79%) due to its completely random decisions. Although Greedy achieves the best results out of all heuristic baselines, its short-sighted optimization leads to significantly higher latencies (2.0-3.5×) and energy consumption (1.9-3.2×) than DHMA. Collectively, these results underscore the inherent limitations of heuristic strategies in adapting to complex and dynamic network conditions.

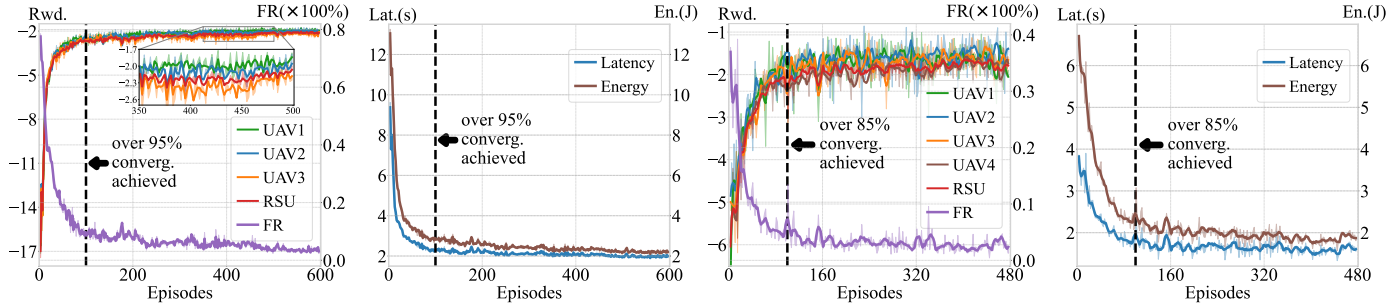
As for DRL-based methods, both M-DRL and TMP-DDPG suffer from suboptimal UAV position scheduling, struggling to alleviate high-demand hot zones effectively. Although DE-TD3 benefits from multi-UAV deployment via Differential Evolution, its lower-level single-agent design introduces decision bottlenecks and local flexibility issues. While AB-MAPPO and MATD3 deliver the strongest results among all DRL-based baselines, they still trail DHMA due to their absence of dynamics prediction, which makes them less responsive to upcoming traffic changes. Compared with AB-MAPPO on four scenarios, DHMA achieves relative improvements of 12.1%, 14.1%, 17.9%, and 17.8% w.r.t. reward.

To further verify the real-time feasibility of DHMA, we quantify the controller-side runtime induced by the trajectory prediction component in Appendix D.2. The reported results show that the prediction process incurs acceptable overhead and the overall online decision runtime stays well below the control interval across all four scenarios.

### 5.2.3 *Ablation Studies*

To comprehensively investigate the contribution of each key design component, we conduct ablation studies from both system modeling and algorithmic perspectives.

- *w/o-u*: removes aerial-level resources provided by UAVs from the system, restricting task processing to ground-based execution on either vehicles or RSUs.
- *w/o-m*: neglects cross-RSU task migration when constructing the system model, disallowing tasks to be redistributed among RSUs once offloaded.
- *w/o-u&m*: simultaneously disables UAV assistance and cross-RSU task migration—two critical mechanisms designed to alleviate spatio-temporal load imbalances.
- *w/o-augA*: deactivates UAV observation augmentation achieved with predicted workload dynamics, limiting the proactive perception of future workload surges.




(a) Reward & FR on $\text{Grid}_{3\times3}$ (b) Latency & Energy on $\text{Grid}_{3\times3}$ (c) Reward & FR on $\text{Net}_4$ (d) Latency & Energy on $\text{Net}_4$

Fig. 7. Convergence performance of our DHMA approach during the centralized training phase on $\text{Grid}_{3\times3}$ and $\text{Net}_4$ scenarios.

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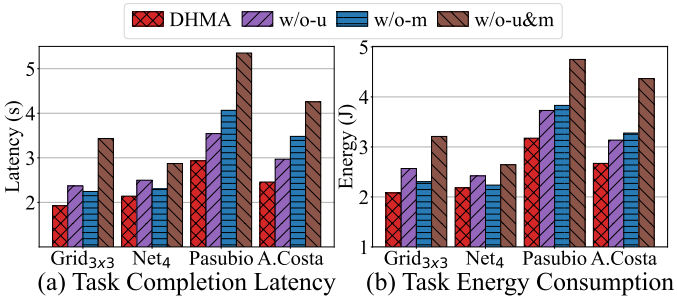



Fig. 8. Latency and energy consumption performance of DHMA variants from the system modeling perspective.

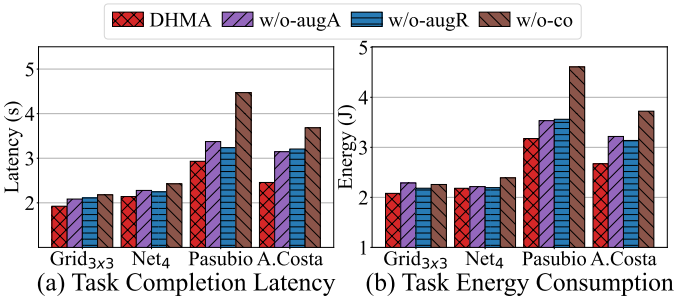



Fig. 9. Latency and energy consumption performance of DHMA variants from the algorithmic perspective.

TABLE 5
Ablation studies on key components of DHMA with Rwd.&FR reported.

| Variants | $\text{Grid}_{3\times3}$ | | $\text{Net}_4$ | | Pasubio | | A.Costa | |
|---|---|---|---|---|---|---|---|---|
| | Rwd. | FR | Rwd. | FR | Rwd. | FR | Rwd. | FR |
| **w/o-u** | -2.526 | 0.101 | -2.977 | 0.077 | -4.880 | 0.254 | -3.627 | 0.226 |
| **w/o-m** | -2.396 | 0.108 | -2.551 | 0.114 | -5.850 | 0.273 | -4.569 | 0.265 |
| **w/o-u&m** | -4.197 | 0.322 | -3.392 | 0.201 | -8.064 | 0.333 | -6.487 | 0.292 |
| **w/o-augA** | -2.143 | 0.052 | -2.512 | 0.093 | -4.610 | 0.247 | -4.382 | 0.238 |
| **w/o-augR** | -2.201 | 0.059 | -2.493 | 0.087 | -4.538 | 0.231 | -4.473 | 0.241 |
| **w/o-co** | -2.376 | 0.092 | -2.837 | 0.128 | -6.074 | 0.285 | -5.016 | 0.274 |
| **DHMA** | **-2.015** | **0.026** | **-2.365** | **0.066** | **-3.799** | **0.211** | **-2.815** | **0.147** |

- *w/o-augR*: deactivates RSU observation augmentation achieved with predicted vehicular movements, hindering anticipatory guidance for cross-RSU task migration.
- *w/o-co*: restricts information exchange across each tier of agents, therefore preventing the sharing of global features.

Table 5 displays the rewards and failure rates of six DHMA variants, and Figures 8 and 9 visualize their latency and energy consumption relative to the complete DHMA. All three variants that disable workload-balancing mechanisms experience considerable performance degradation, among which *w/o-u&m* performs the worst. These indicate that UAV assistance and cross-RSU task migration are jointly essential for responsive and robust workload balancing. Moreover, both *w/o-augA* and *w/o-augR* suffer from notable increases in failure rates—1.2-2.0× and 1.1-2.3× those of DHMA on four scenarios—underscoring the necessity for proactive dynamics prediction in guiding timely and collaborative decision-making. By restricting inter-agent information exchange, *w/o-co* forces agents to operate independently without collaboration, further worsening the average |Rwd.| and FR by 1.44× and 2.17× compared with DHMA.

### 5.2.4 Case Studies

To evaluate the effectiveness of our workload-balancing mechanisms, we visualize the workload distributions of several representative approaches on Pasubio scenario, as shown in Figure 10. The workload of each RSU is measured by the number of executed tasks, and the overall balancing is quantified with Jain's Fairness Index (JFI).

We can observe that RSU-O exhibits significant load imbalances with concentrated hot zones at multiple high-traffic intersections, while PO-50% alleviates these to some extent but remains incapable of adapting to dynamic variations in traffic volume. Although w/o-u&m reduces the severity of load imbalances, it still suffers from noticeable task accumulation in overloaded RSUs (e.g. 309.2 in orange) while

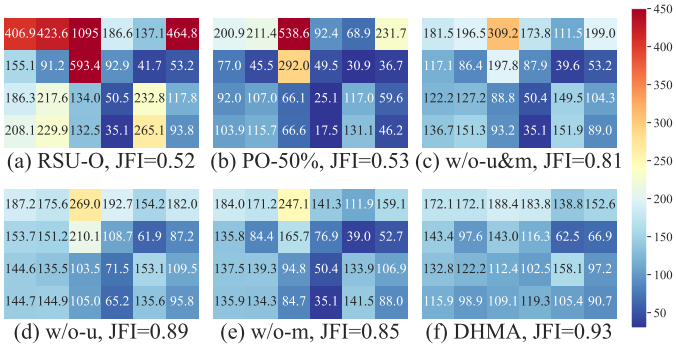



Fig. 10. RSU workload distributions of representative approaches on Pasubio scenario. Each grid corresponds to an intersection with colors indicating workload levels (red = overloaded, blue = underloaded).

leaving others underutilized (e.g. 35.1 in dark blue). When at least one mechanism is retained, as in w/o-u and w/o-m, the workloads become more evenly distributed, underscoring the indispensable roles of UAV assistance and cross-RSU task migration in enhancing computation offloading. In contrast, DHMA achieves the most uniform system-wide workload balancing (JFI=0.93) with no prominent hot zones. This is enabled by its effective aerial-terrestrial cooperation, which jointly exploits UAV supplementary resources and dynamic cross-RSU task migration with the guidance of predictive observation augmentation at both tiers.

### 5.2.5 Stress Tests

We further validate the robustness of DHMA under prediction mismatch through two end-to-end stress tests on $\text{Net}_4$ and A.Costa without retraining the learned policies. **(i) Prediction Perturbation.** We inject controlled errors into the predicted trajectories as $\tilde{\mathcal{X}}_{i,j}^{t+l} = (1-\eta)\hat{\mathcal{X}}_{i,j}^{t+l} + \eta\hat{\mathcal{X}}_{\pi(i,j)}^{t+l}$, where $\eta$ is swept from 0 to 0.5. Future workload profiles and vehicle–RSU association sequences are then regenerated from the perturbed trajectories. **(ii) Unexpected Vehicle Deviation.** We allow vehicles to switch to an alternative feasible route with probability $p_m \in \{0, 0.05, 0.10, 0.15, 0.20\}$ while keeping the predicted trajectories unchanged. We compare DHMA with Oracle-DHMA, *w/o-augA*, and *w/o-augR* under varying degrees of mismatch. As shown in Figure 11, DHMA degrades gracefully towards *w/o-augA* and *w/o-augR* as $\eta$ or $p_m$ increases, rather than suffering abrupt performance collapse. Oracle-DHMA remains the upper bound, confirming that predictive observation augmentation improves decision quality while preserving robustness under prediction errors and sudden mobility deviations.

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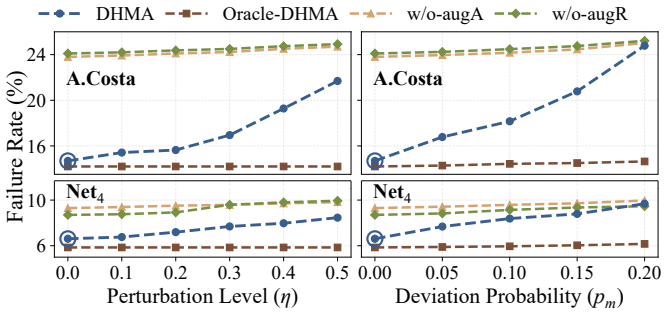
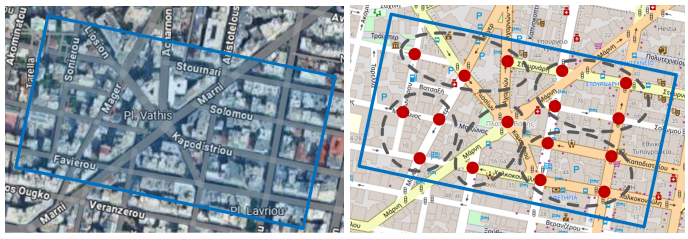



Fig. 11. Robustness of DHMA under prediction perturbation and unexpected vehicle deviation on $\text{Net}_4$ and A.Costa scenarios.



(a) Satellite Imagery (b) Road Network Topology

Fig. 12. Real-world visualization of Region 9 in Athens, Greece. (a) Satellite imagery from Google Maps (https://www.google.com/maps) showing the urban layout. (b) Road network topology annotated with 16 RSUs (red dots) and 5 UAVs (dashed circles).

### 5.2.6 Real-World Validation

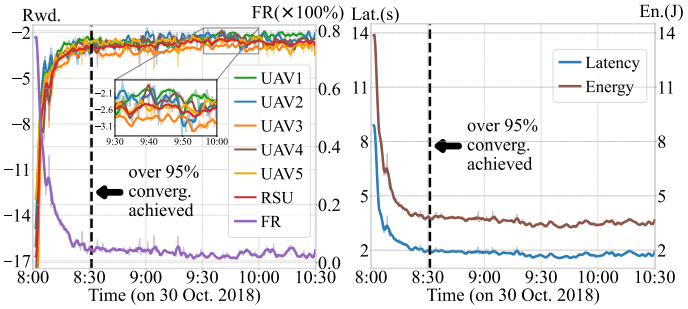
To assess the generalization and practical viability of our DHMA approach, we conduct additional experiments using the open-source naturalistic trajectory dataset pNEUMA [56], collected in the congested downtown area of Athens, Greece. Specifically, we extract the trajectory records of Region 9 (blue rectangle in Figure 12a) during 8:00-10:30 on 30 Oct. 2018, and construct a corresponding real-world scenario with 16 RSUs and 5 UAVs as depicted in Figure 12b. A total of 4,362 car trajectories are downsampled to a temporal resolution of 0.2 s and divided into 10-second episodes, each comprising $T = 50$ time slots of length 0.2 s. The computing powers of UAVs and RSUs are set to $f_k^A = 10$ GHz and $f_i^R = 18$ GHz, and the data transmission rate $\beta$ is set to 1 Gbits/s. Each vehicle generates one computation task per time slot. As shown in Figure 13, DHMA achieves stable convergence within the first 30 minutes (by 8:30), reaching over 95% of its final performance across all agents.

**Realistic Wireless Conditions.** We consider an extended channel setting with explicit LoS/NLoS propagation and small-scale fading; the detailed formulations are provided in Appendix D.3. We tune the local computing power $u_{i,j}^t$ within the range of $[2, 3]$ GHz. The results summarized in Table 6 reveal that, as $u_{i,j}^t$ increases, the failure rates of all approaches decrease due to enhanced local processing capability. More importantly, although the harsher channel conditions degrade the performance of all approaches, DHMA still outperforms all baselines under all tested configurations. These results indicate that the effectiveness of DHMA is not tied to the simplified quasi-static channel abstraction, and validate its strong adaptability and robustness in real-world traffic and wireless environments.

**Impact of UAV Deployment Scale.** We vary the number of deployed UAVs and repartition the 16 RSUs into disjoint

TABLE 6
Task failure rates under varying vehicle computing powers on real-world Athens scenario with extended channel setting.

| Approach | Computing Power $u_{i,j}^t$ (GHz) | | | | |
|---|---|---|---|---|---|
| | 2.00 | 2.25 | 2.50 | 2.75 | 3.00 |
| PO-50% | 0.5307 | 0.4202 | 0.3193 | 0.2677 | 0.2428 |
| Greedy | 0.4908 | 0.3680 | 0.2788 | 0.2154 | 0.1991 |
| M-DRL | 0.4329 | 0.2985 | 0.2193 | 0.1441 | 0.1296 |
| DE-TD3 | 0.3705 | 0.2391 | 0.1572 | 0.0756 | 0.0678 |
| AB-MAPPO | 0.3213 | 0.2070 | 0.1375 | 0.0487 | 0.0215 |
| DHMA | **0.2972** | **0.2031** | **0.1239** | **0.0254** | **0.0085** |




(a) Reward & FR on Athens (b) Latency & Energy on Athens

Fig. 13. Convergence performance of DHMA during real-world validation in Athens, Greece from 8:00 to 10:30 on 30 Oct. 2018.

UAV service regions, while keeping all other settings unchanged with $u_{i,j}^t$ fixed at 2.5 GHz. As shown in Table 7, increasing UAVs improves the average transmission rate and overall service performance, since more aerial-level resources can be flexibly dispatched to overloaded RSUs for surge absorption. Meanwhile, the performance gain gradually decreases as the number of UAVs increases from 7 to 9, indicating diminishing marginal returns once the major hot zones have been effectively alleviated.

TABLE 7
Performance under different numbers of UAVs on real-world Athens scenario with $u_{i,j}^t = 2.5$ GHz.

| #UAVs | Rate(Mbps) | Lat.(s) | En.(J) | FR |
|---|---|---|---|---|
| 1 | 35.2721 | 2.6127 | 3.4086 | 0.1718 |
| 3 | 41.6852 | 2.4861 | 3.2533 | 0.1592 |
| 5 | 45.9388 | 2.2048 | 2.9701 | 0.1239 |
| 7 | 47.3192 | 2.0891 | 2.8723 | 0.1064 |
| 9 | 47.9761 | 2.0418 | 2.8354 | 0.0986 |

## 6 Conclusion

In this work, we investigate a UAV-assisted multi-edge collaborative computation offloading problem in vehicular networks. By incorporating UAV assistance and enabling cross-RSU task migration, we aim to achieve both responsive and robust workload balancing through effective aerial-terrestrial cooperation. We propose a Dynamics-Aware Hierarchical Multi-Agent deep reinforcement learning approach, and design a two-tiered decision framework to jointly optimize UAV position scheduling and vehicular computation offloading. In addition, a dynamics-aware observation augmentation method is developed to guide the timely and collaborative decision-making of agents with predicted workload surges and vehicular movements. Extensive experiments across four simulation scenarios validate the superiority of our approach. Besides, our ablation studies, case

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studies, and real-world validation further demonstrate its robustness and practical viability.

For future work, we plan to relax our idealized wireless-coverage assumption by incorporating more realistic wireless communication characteristics, such as irregular or overlapping coverage footprints, mobility-induced handovers, and transient link interruptions. We will further enrich the system model by explicitly incorporating UAV flight time and propulsion energy, enabling fine-grained continuous UAV trajectory planning under delay and battery constraints. In addition, we will integrate security, privacy, and trust considerations to enable authenticated, privacy-preserving, and auditable coordination among devices.

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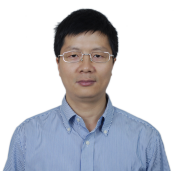
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